

DPW-8/AePW-4 Scatter Working Group:

An Overview of Mini Workshops 1 and 2



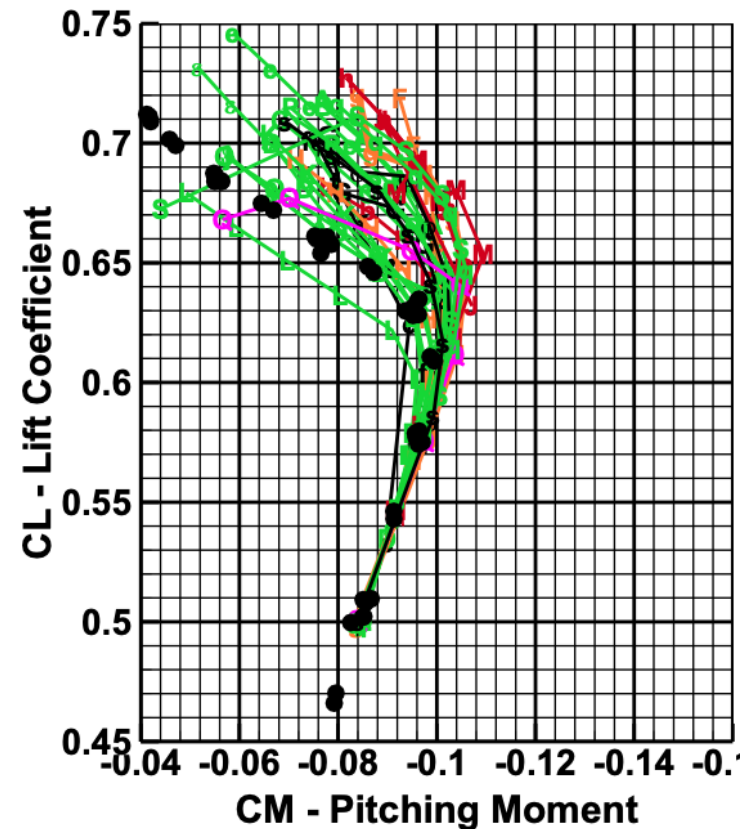
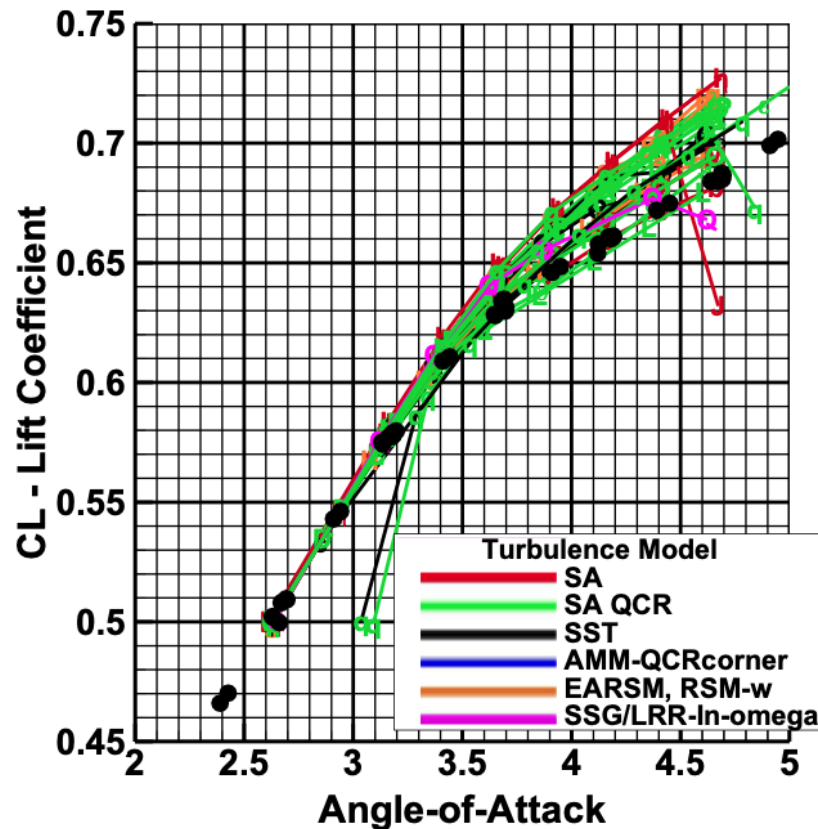
AIAA Aviation 2025
July 21

<https://www.aiaa-dpw.org>



Source of Scatter – Motivation

- Seek to identify deviations in DPW-7 CRM data
- Significant spread in solvers post pitchup (all submissions plotted)



Curves collapsed to match experimental data near cruise point

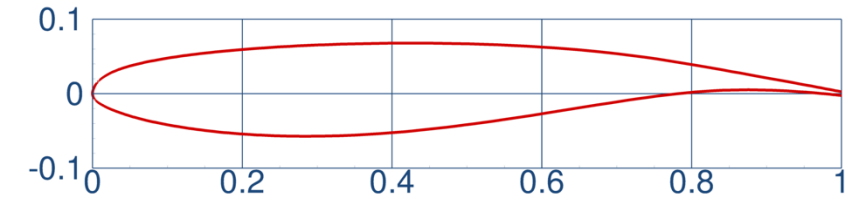
Mach = 0.85, Re=20M - CFD Shifted to Match Test at $C_L=0.53$

Image source:
Tinoco, E., et al., "Summary Data from the Seventh AIAA CFD Drag Prediction Workshop," AIAA 2023-3492

- **Three test cases examined so far**
 - Test Case 1a: ONERA OAT15A
 - Establish initial level of scatter
 - Test Case 1b: Joukowski Airfoil
 - Order of accuracy check
 - Test Case 1c: ONERA OAT15A
 - Reduced scatter
- **Future test case**
 - Test Case 2: CRM Wing/Body Cruise
 - Examine scatter for 3D and QCR2000
- **Sustained meeting cadence and structure**
 - Approx 20 people on distribution list
 - Average 10 attendees in each meeting
 - Meeting Tuesdays 10am ET on 2nd and 4th week of the month

Test Case 1a: Workshop-Wide Validation

- **Validation of steady CFD analysis, required**
- **Users are encouraged to employ best practices**
- **Settings**
 - Steady CFD (e.g., RANS)
 - Prefer some version of SA, multiple turbulence models can be submitted
 - Purely 2D simulations (one cell wide)
- **Grids**
 - Six-member RANS grid family; four are required, six are desirable
 - Encourage use of committee-supplied grids; user-generated grids are acceptable
 - Committee-supplied grid is one cell wide with a 230mm chord (same as experiment) and follows RANS best practices
- **Conditions**
 - Mach 0.73, $Re_c=3m$ (based on chord length), $T_{static}=271\text{ K (487.8 R)}$
 - Alpha: 1.36, 1.50, 2.50, 3.00, 3.10



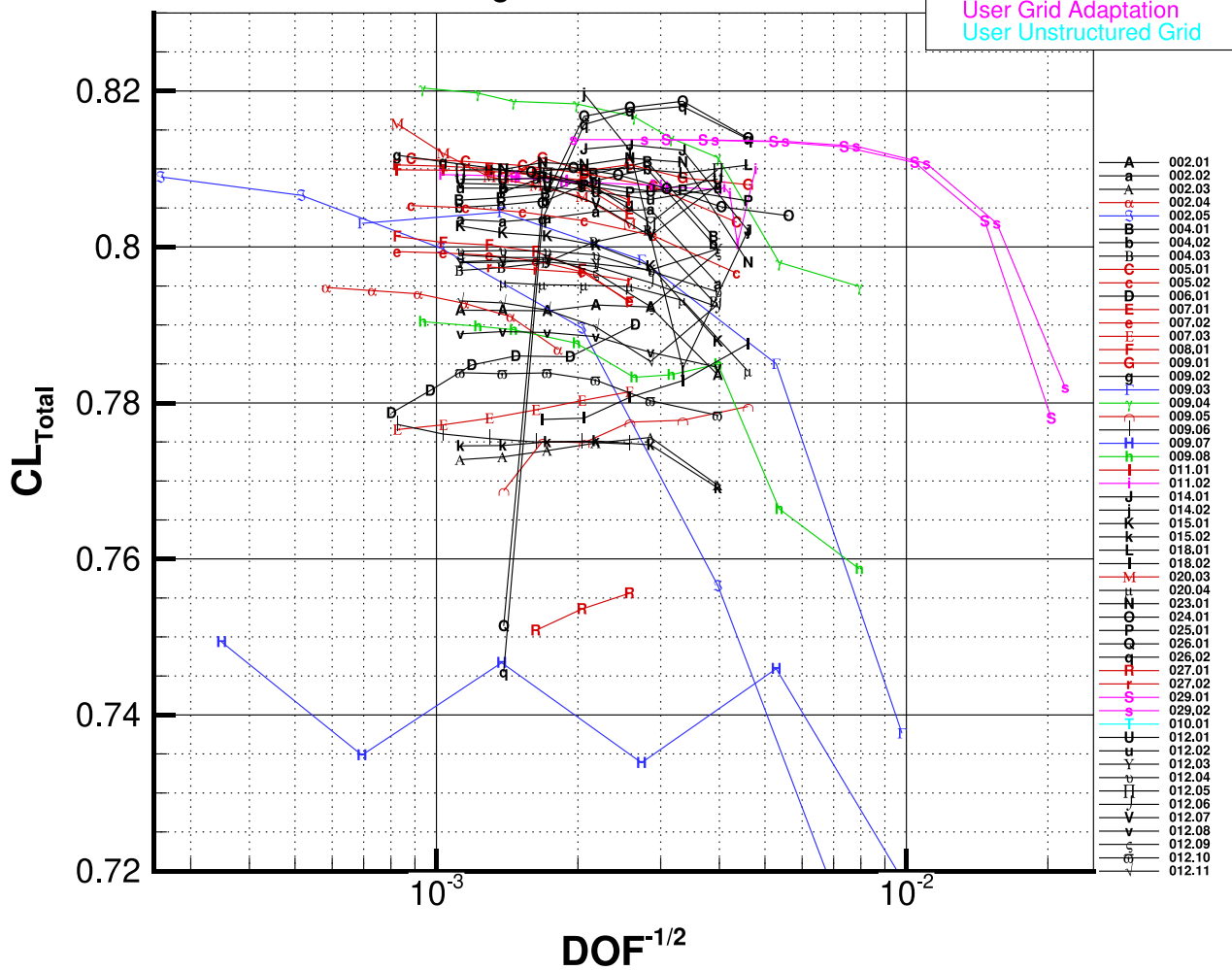
ONERA OAT15A Transonic Airfoil

Jaquin, et al. "Experimental Study of Shock Oscillation over a Transonic Supercritical Profiles." AIAA Journal, Vol. 47, No. 9, 2009. Pages 1985-1994.

ONERA OAT15A C_L Convergence: $\alpha = 1.5^\circ$

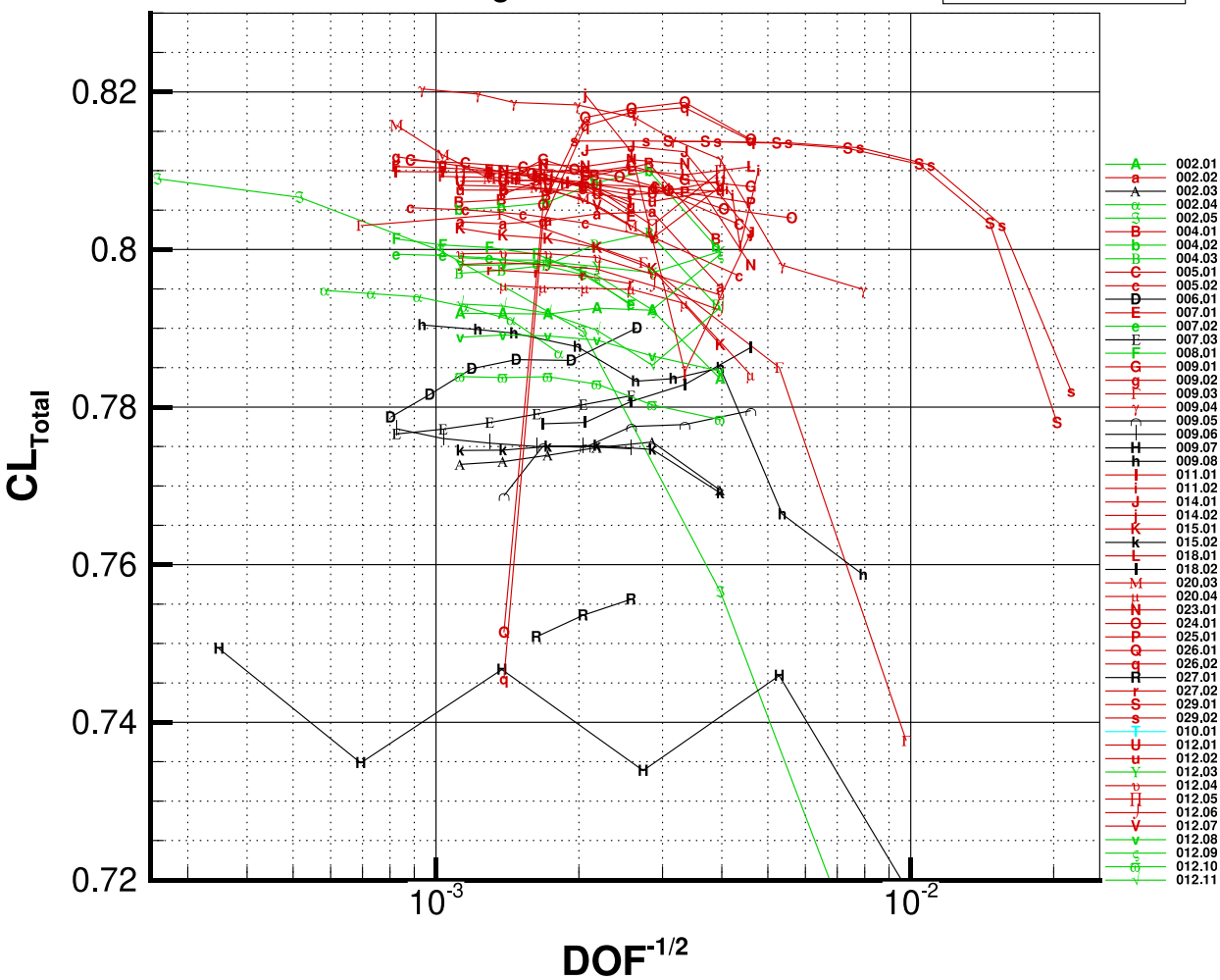
Test Case 1a: ONERA OAT15A
Grid Convergence Study
Angle of Attack 1.50

Grid Family:
Cadence Structured
Cadence Unstructured
HeldenMesh Unstructured
ONERA Structured
User Grid Adaptation
User Unstructured Grid

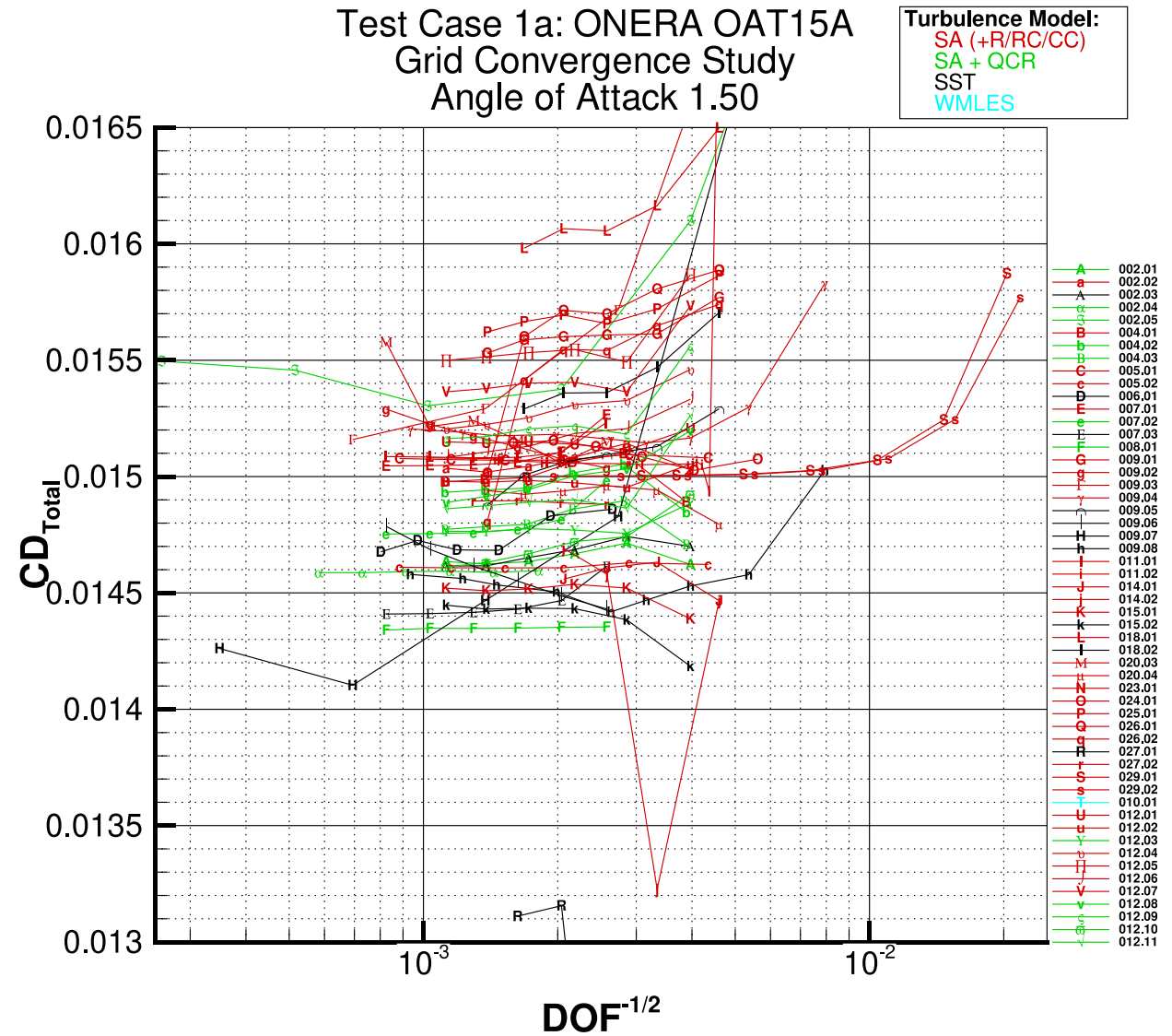
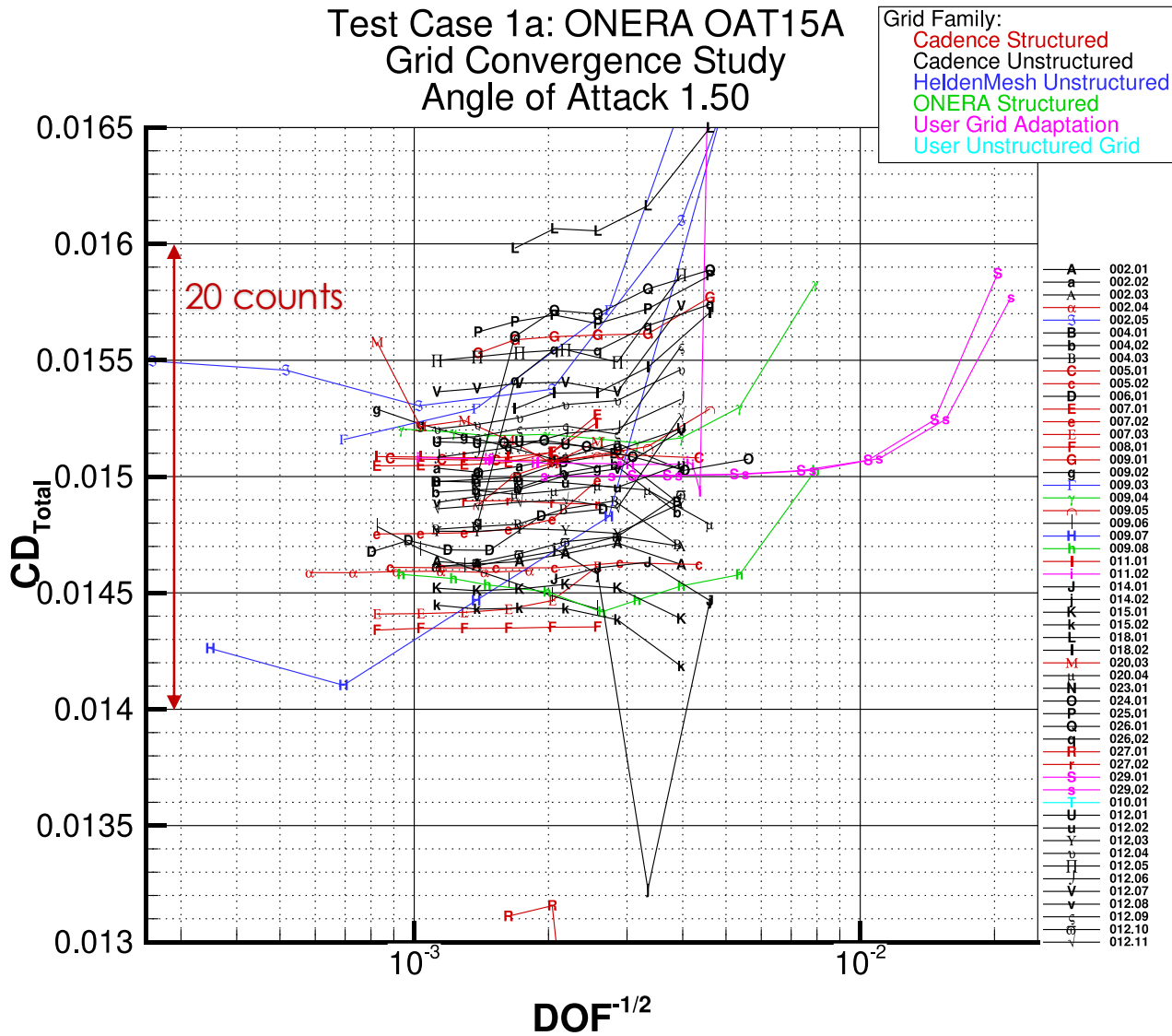


Test Case 1a: ONERA OAT15A
Grid Convergence Study
Angle of Attack 1.50

Turbulence Model:
SA (+R/RC/CC)
SA + QCR
SST
WMLES



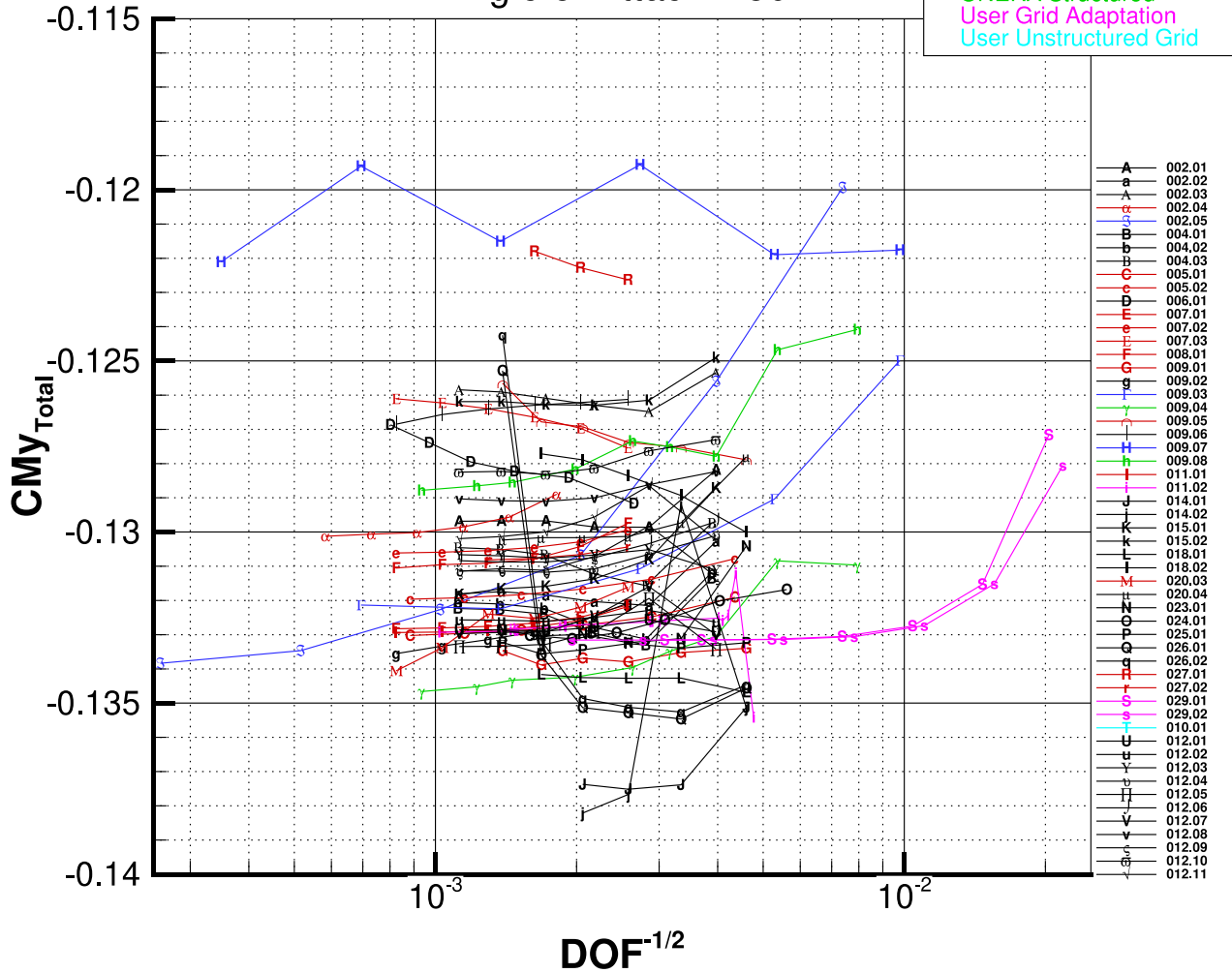
ONERA OAT15A C_D Convergence: $\alpha = 1.5^\circ$



ONERA OAT15A C_M Convergence: $\alpha = 1.5^\circ$

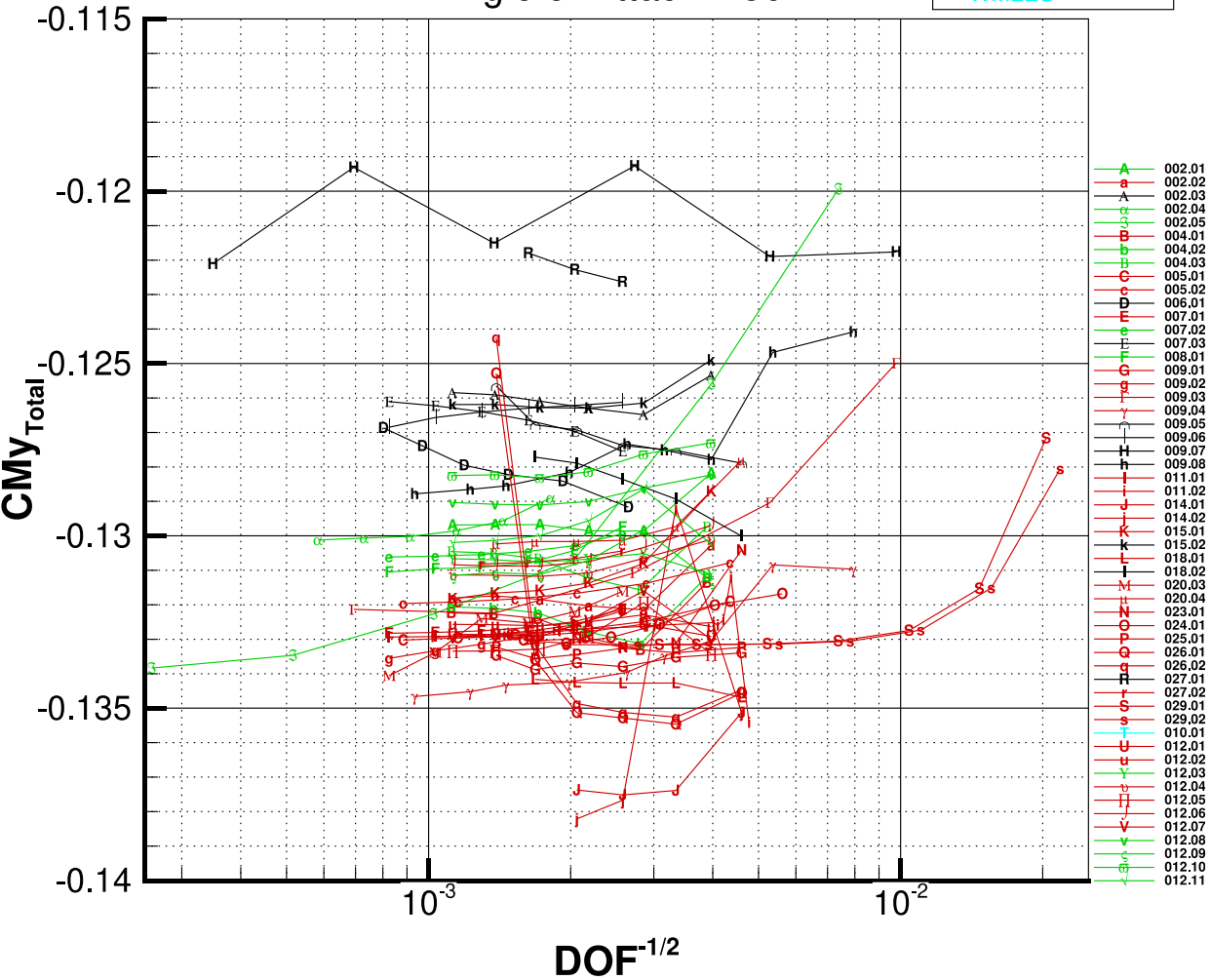
Test Case 1a: ONERA OAT15A
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Grid Family:
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ONERA Structured
User Grid Adaptation
User Unstructured Grid

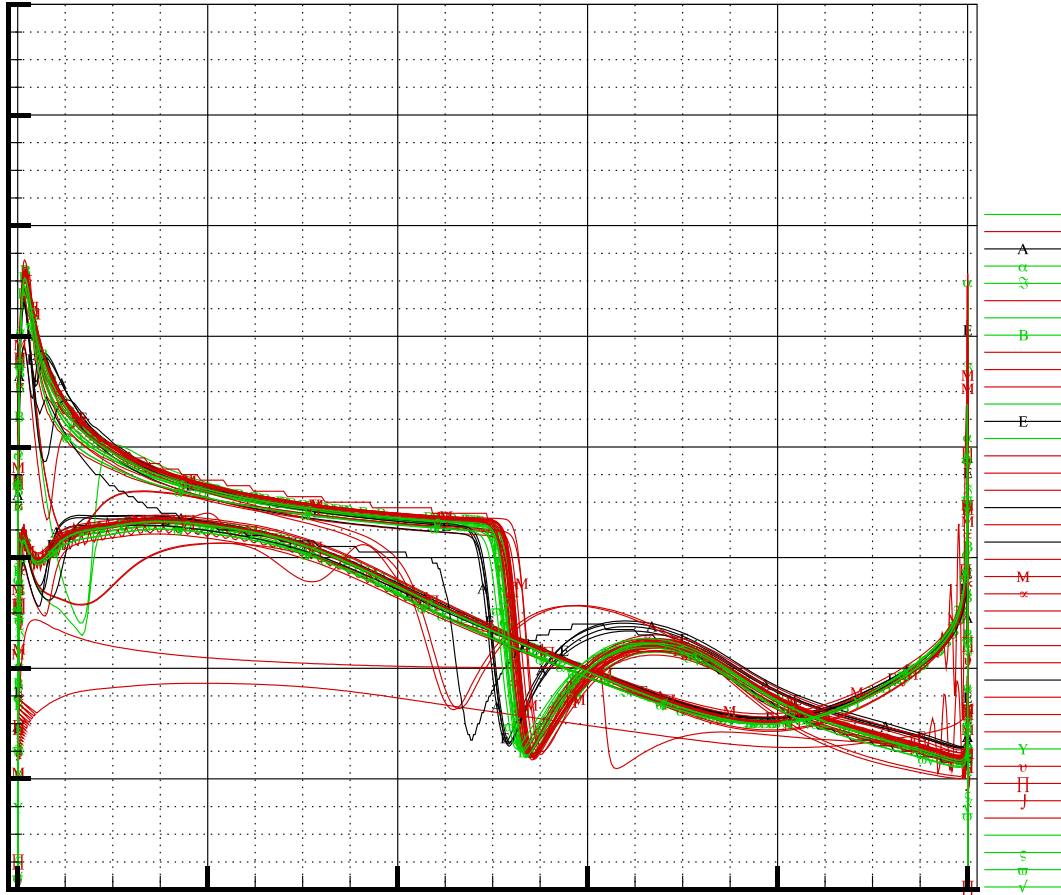
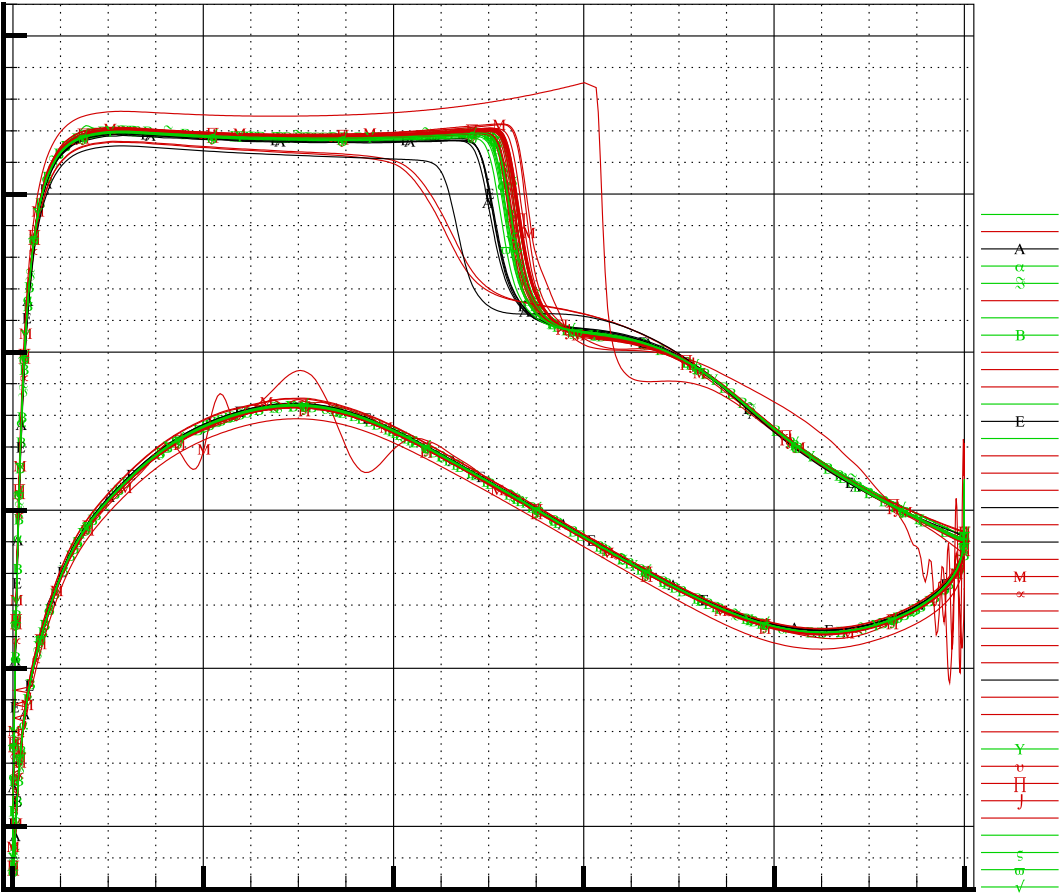


Test Case 1a: ONERA OAT15A
Grid Convergence Study
Angle of Attack 1.50

Turbulence Model:
SA (+R/RC/CC)
SA + QCR
SST
WMLES



ONERA OAT15A C_p and C_f : $\alpha = 1.5^\circ$



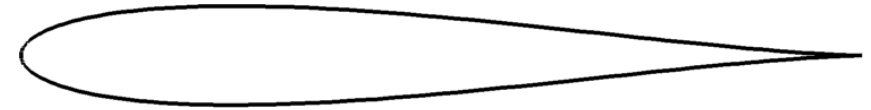
Test Case 1b: Joukowski Airfoil

- **Validation of steady CFD analysis, required**

- <https://github.com/Drag-Prediction-Workshop/DPW8-Scatter/blob/main/TestCase1b/Joukowski.pdf>

- **Settings**

- Steady CFD RANS **French Vanilla SA-[neg] (All terms!)**
 - Adiabatic Wall (not isothermal)
 - Characteristic Farfield (**100 chords away - no circulation**)
 - Use periodic boundary conditions for sidewall boundary conditions
 - Converge residuals to machine precision (**~1e-10**)



- **Grids**

- Committee-supplied grid family (High-Fidelity CFD Verification Workshop 2024)

- **Conditions**

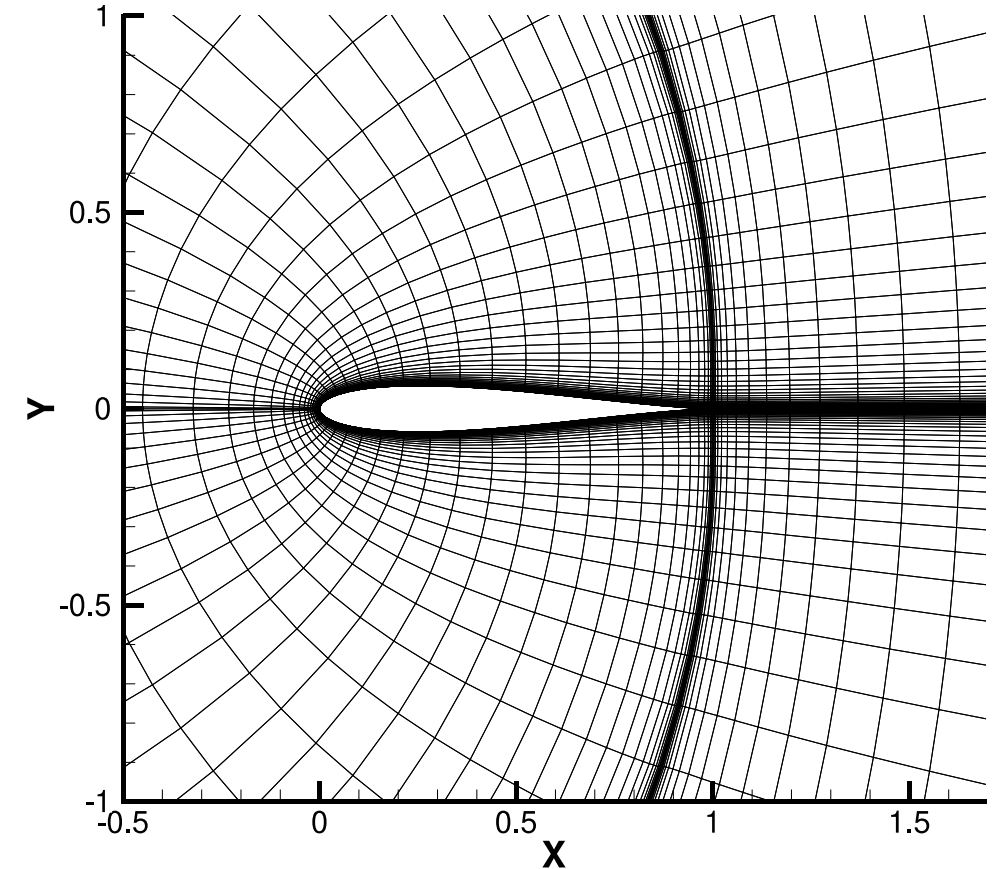
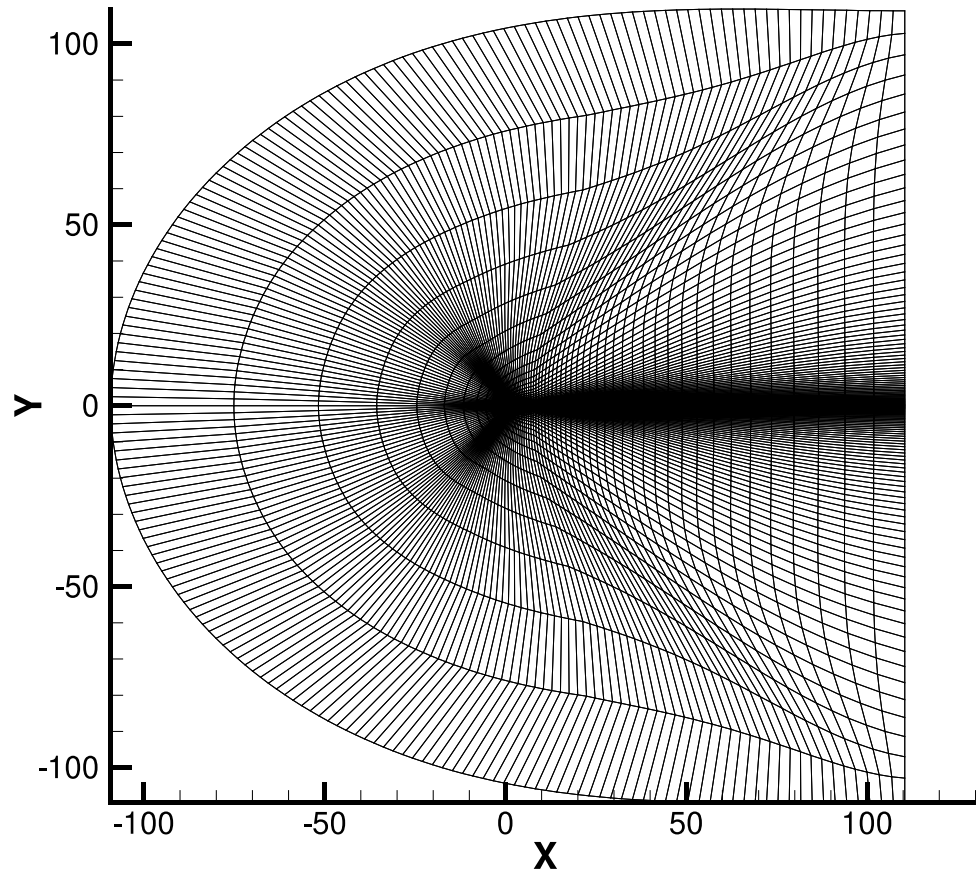
Mach	Re _c	T _{static}	α	γ	Pr	Pr _t	Farfield $\chi = \tilde{\nu}/\nu$
0.15	6×10^6	520.0 R	0.0°	1.4	0.72	0.9	3

- **Sutherland's Law**

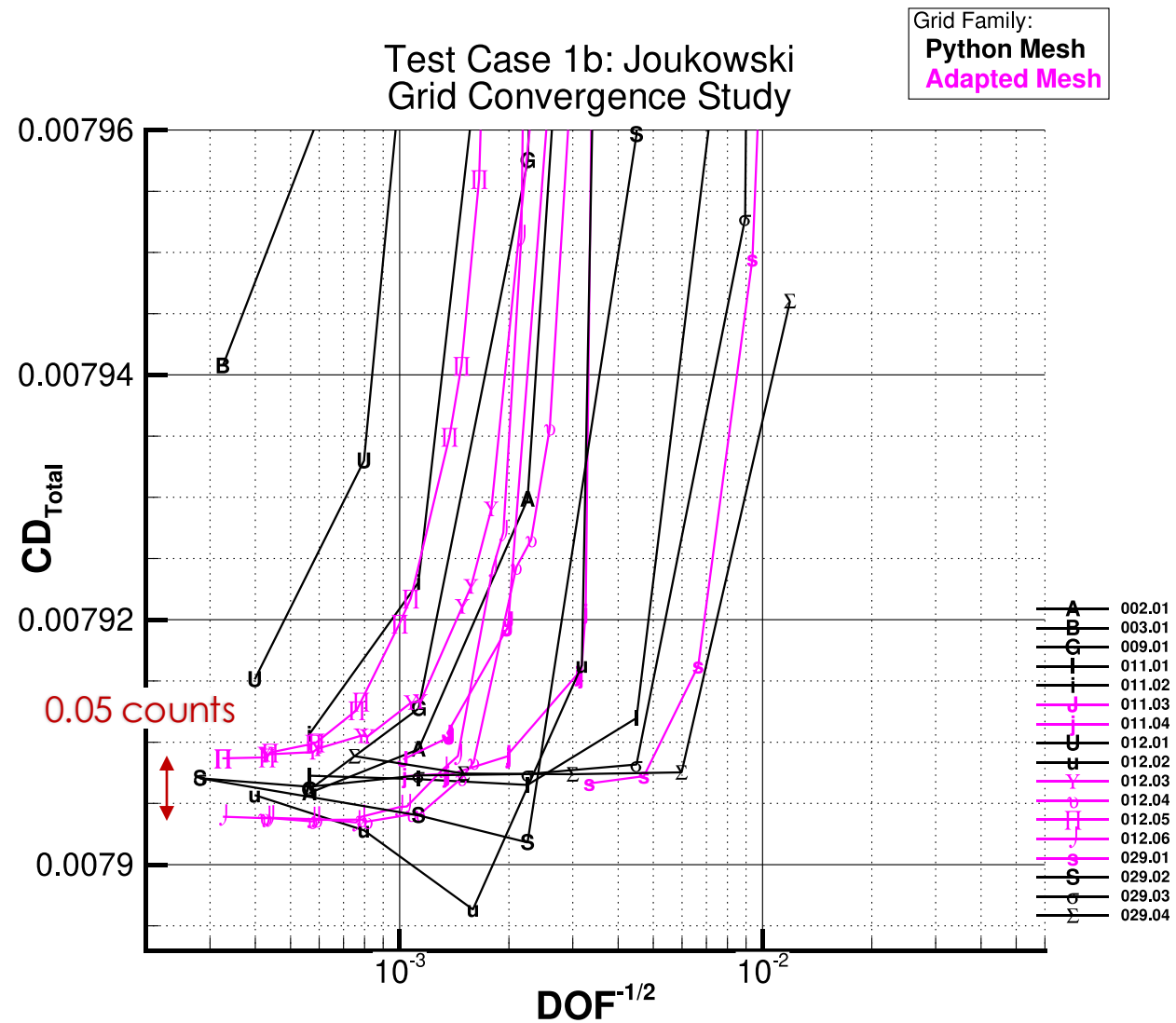
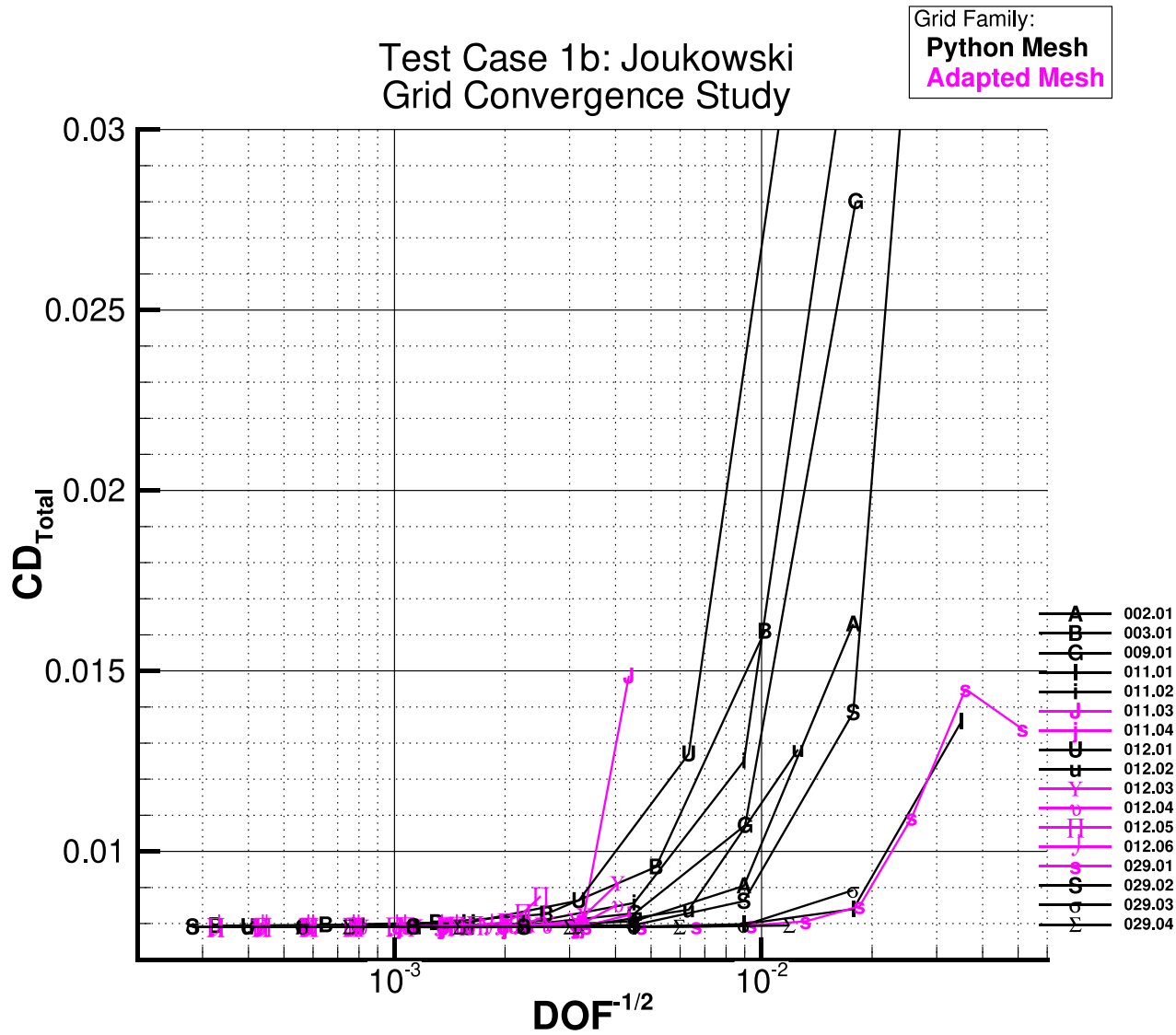
$$\mu(T) = \mu_0 \left(\frac{T}{T_0} \right)^{3/2} \left(\frac{T_0 + S}{T + S} \right) \quad \begin{matrix} \mu_0 = 1.716 \times 10^{-5} \frac{\text{kg}}{\text{m s}} \\ T_0 = 491.6^\circ \text{R} \\ S = 198.6^\circ \text{R} \end{matrix} \quad \frac{\mu(T)}{\mu_{ref}} = \left(\frac{T}{T_{ref}} \right)^{3/2} \left(\frac{1 + S/T_{fef}}{T/T_{fef} + S/T_{fef}} \right)$$

Test Case 1b: Joukowski Airfoil Mesh

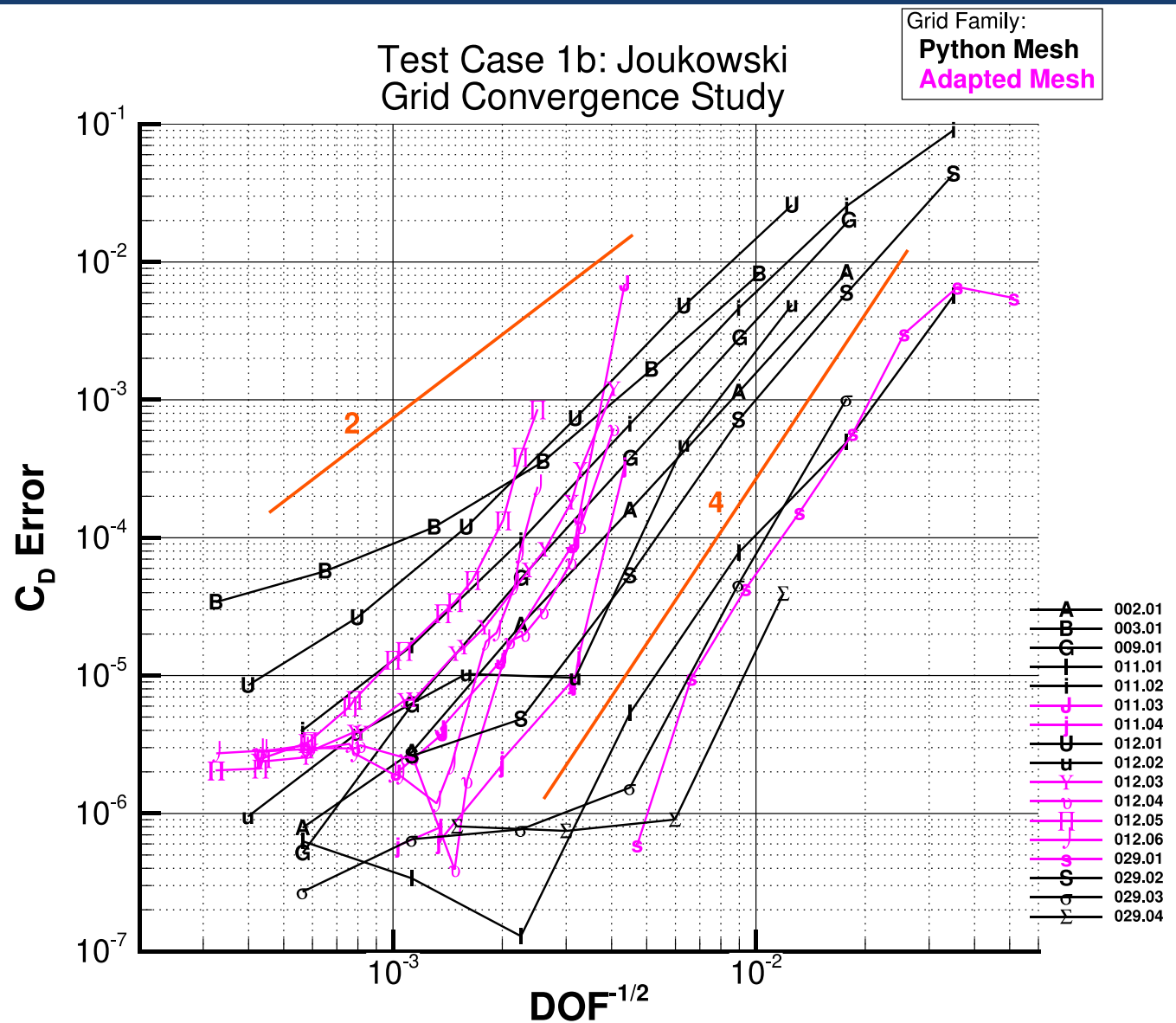
- Cusped trailing edge – remove inviscid singularity
- Zero angle of attack – stagnation point at leading edge
- Custom mesh to observe order of accuracy (Joukowski conformal mapping)



Joukowski C_D Convergence



Joukowski Airfoil – Order of Accuracy



Test Case 1c: ONERA OAT15A Airfoil

- **Verification of steady CFD analysis, required**

- **Settings**

- Steady CFD RANS **French Vanilla SA-[neg] (All terms!)**
 - Adiabatic Wall (not isothermal)
 - Characteristic Farfield (**1000 chords away**)
 - Use periodic boundary conditions for sidewall boundary conditions
- Converge residuals to machine precision (**~1e-10**)

- **Grids**

- Six-member grid family; four are required, six are desirable
- Encourage use of committee-supplied grids; user-generated grids are acceptable
 - Cadence **Structured**/Unstructured, **Helden Mesh** Unstructured

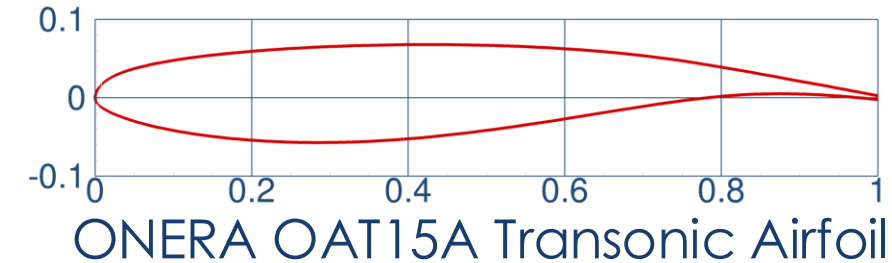
- **Conditions**

Mach	Re _c	T _{static}	α	γ	Pr	Pr _t	Farfield $\chi = \tilde{\nu}/\nu$
0.73	3×10^6	271 K (487.8 R)	1.5°	1.4	0.72	0.9	3

- **Sutherland's Law**

$$\mu(T) = \mu_0 \left(\frac{T}{T_0} \right)^{3/2} \left(\frac{T_0 + S}{T + S} \right) \quad \begin{matrix} \mu_0 = 1.716 \times 10^{-5} \frac{\text{kg}}{\text{m s}} \\ T_0 = 491.6^\circ \text{R} \\ S = 198.6^\circ \text{R} \end{matrix} \quad \frac{\mu(T)}{\mu_{ref}} = \left(\frac{T}{T_{ref}} \right)^{3/2} \left(\frac{1 + S/T_{fef}}{T/T_{fef} + S/T_{fef}} \right)$$

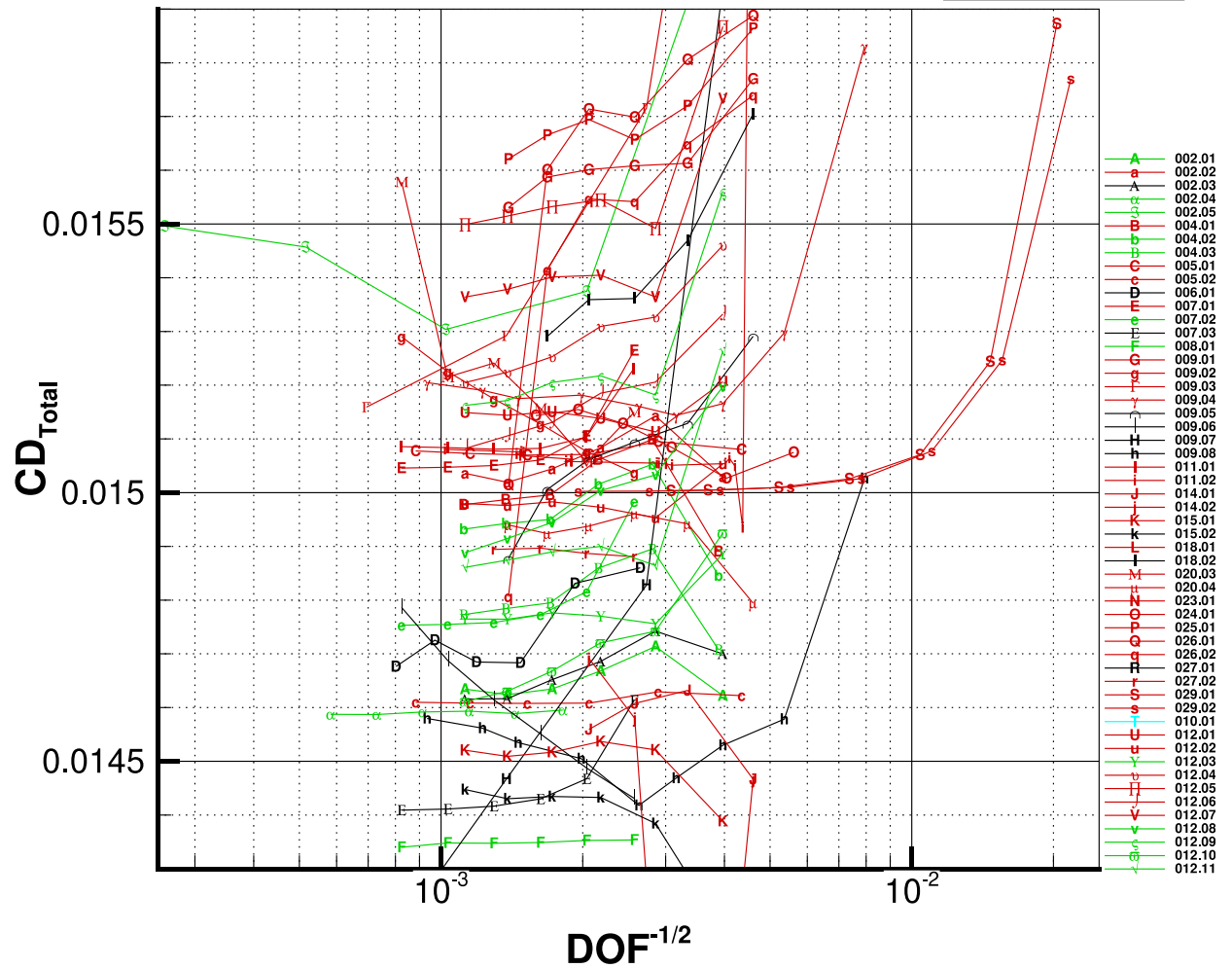
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ONERA OAT15A C_D Convergence: $\alpha = 1.5^\circ$

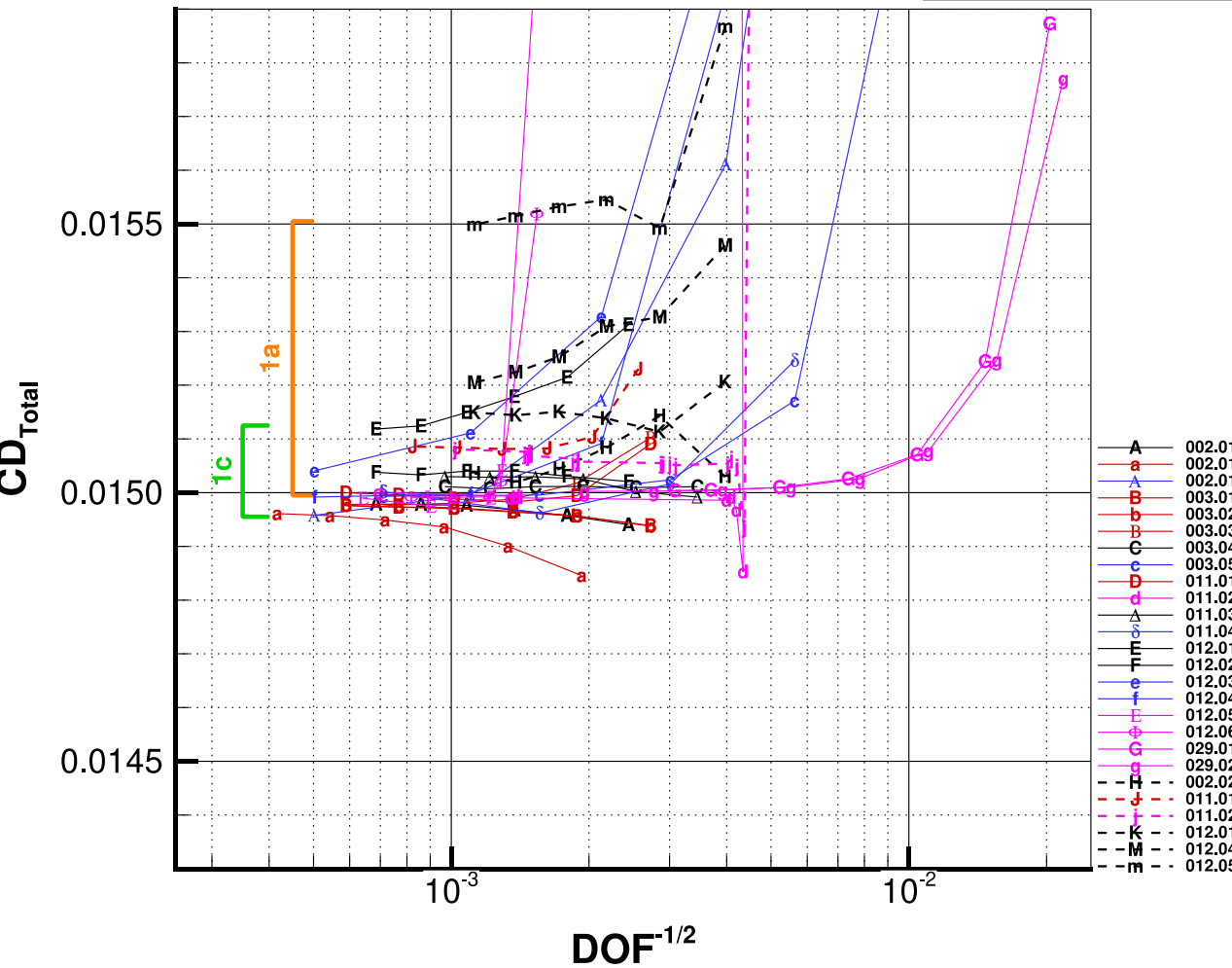
Test Case 1a: ONERA OAT15A
Grid Convergence Study
Angle of Attack 1.50

Turbulence Model:
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SA + QCR
SST
WMLES



Test Case 1c: ONERA OAT15A
Grid Convergence Study
Angle of Attack 1.50

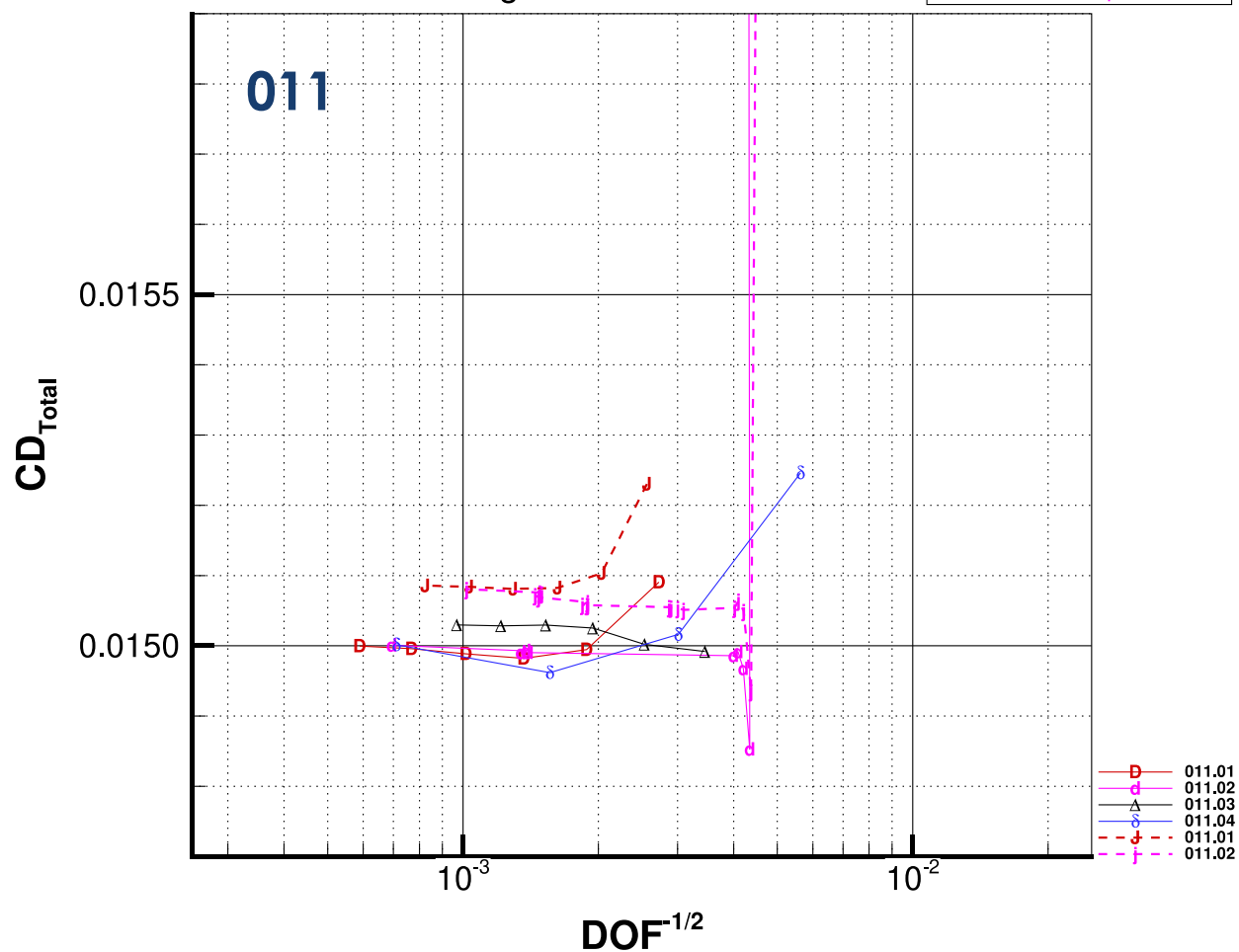
Grid Family:
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Cadence Unstructured
HeldenMesh Unstructured
User Grid Adaptation



ONERA OAT15A C_D Participant Improvements

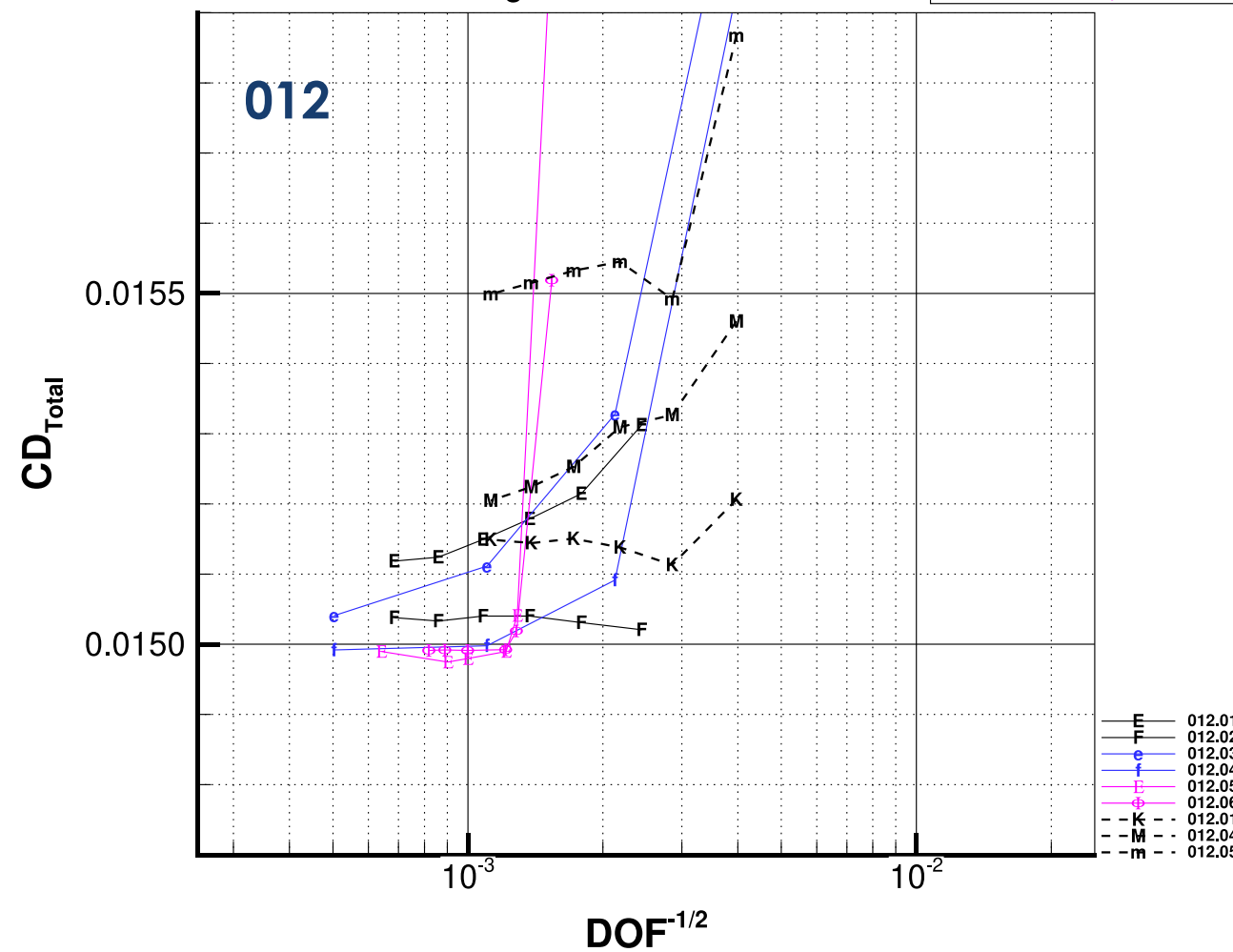
Test Case 1c: ONERA OAT15A
Grid Convergence Study
Angle of Attack 1.50

Grid Family:
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HeldenMesh Unstructured
User Grid Adaptation

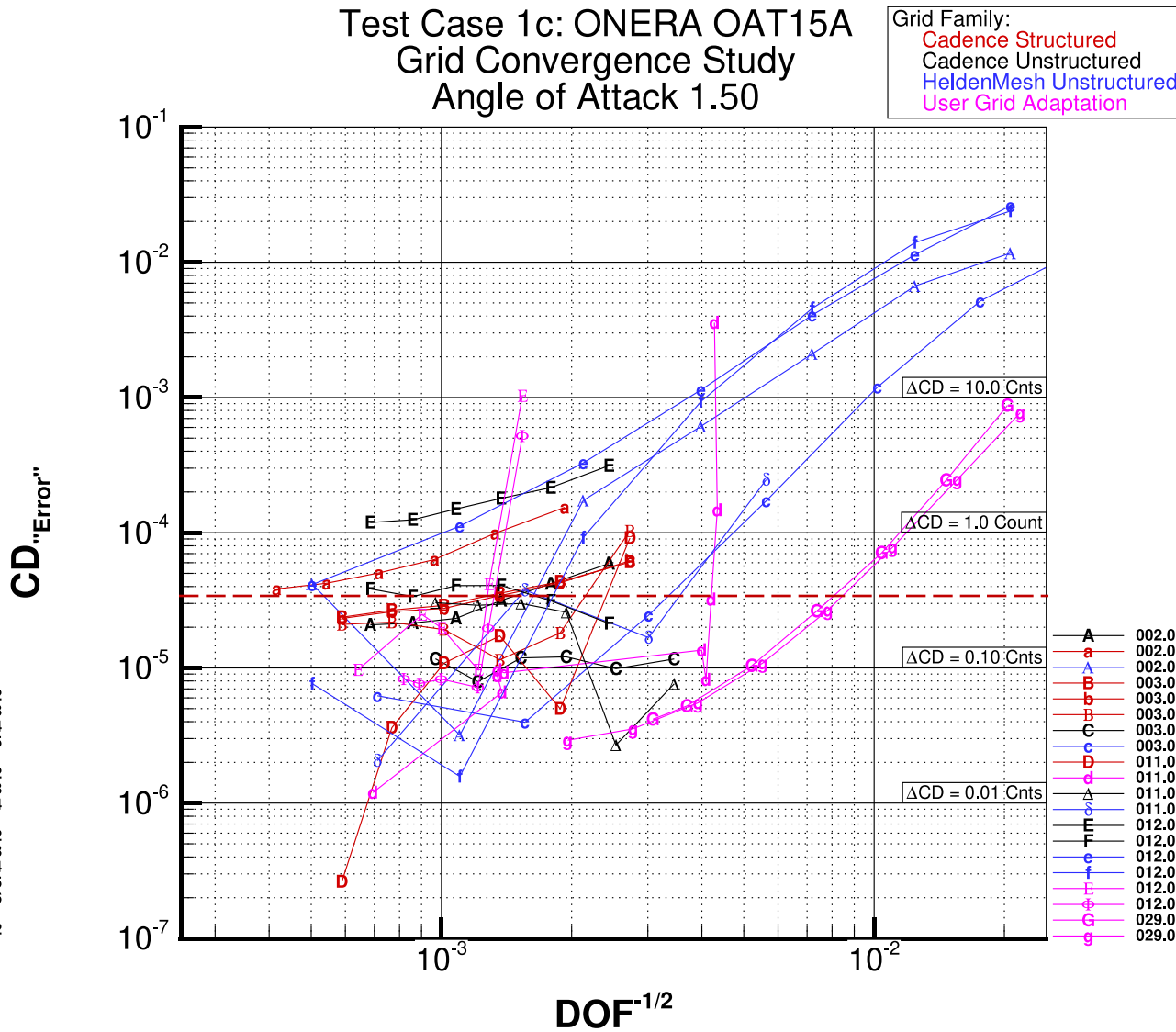
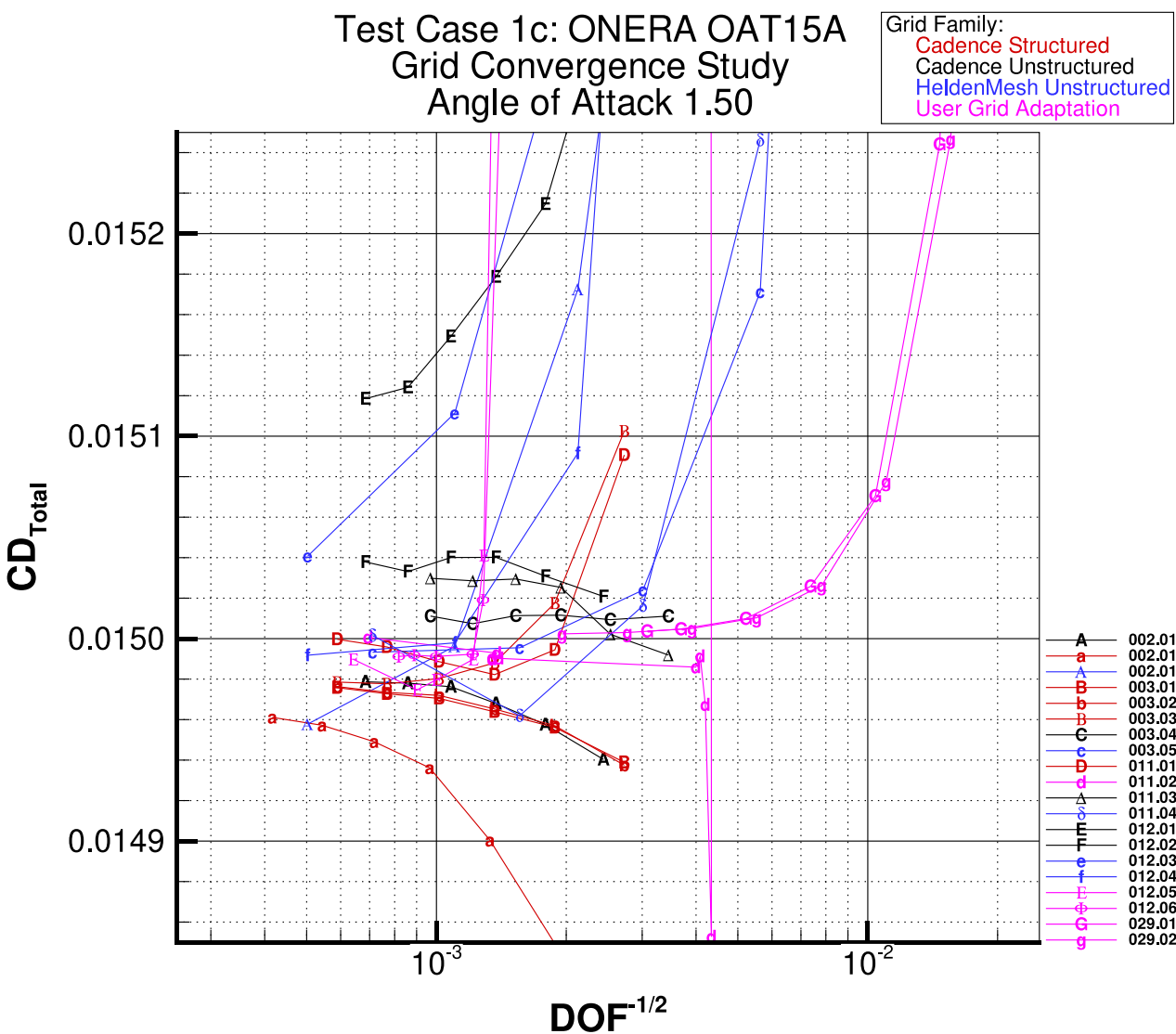


Test Case 1c: ONERA OAT15A
Grid Convergence Study
Angle of Attack 1.50

Grid Family:
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HeldenMesh Unstructured
User Grid Adaptation



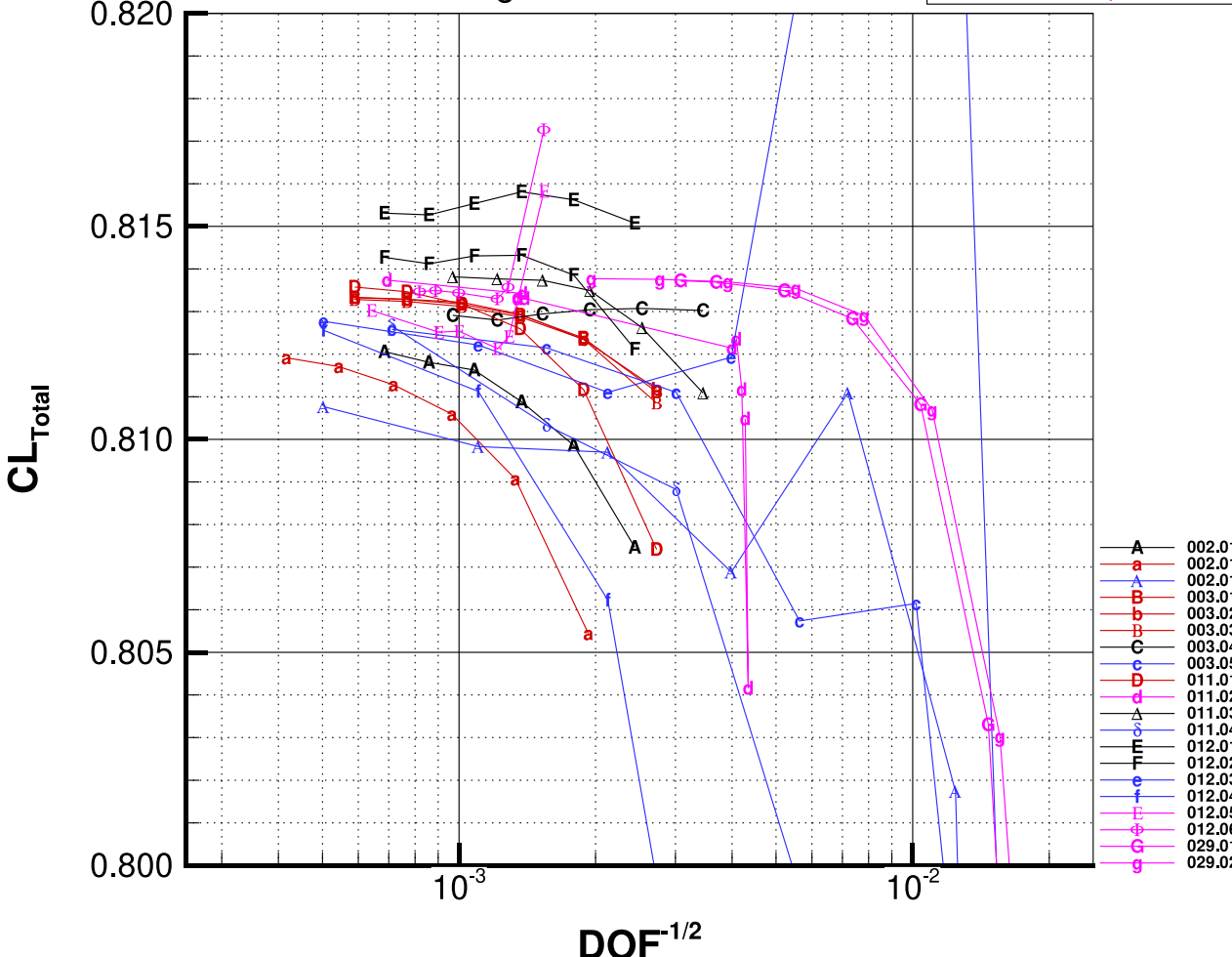
ONERA OAT15A C_D Convergence: $\alpha = 1.5^\circ$



ONERA OAT15A C_L Convergence: $\alpha = 1.5^\circ$

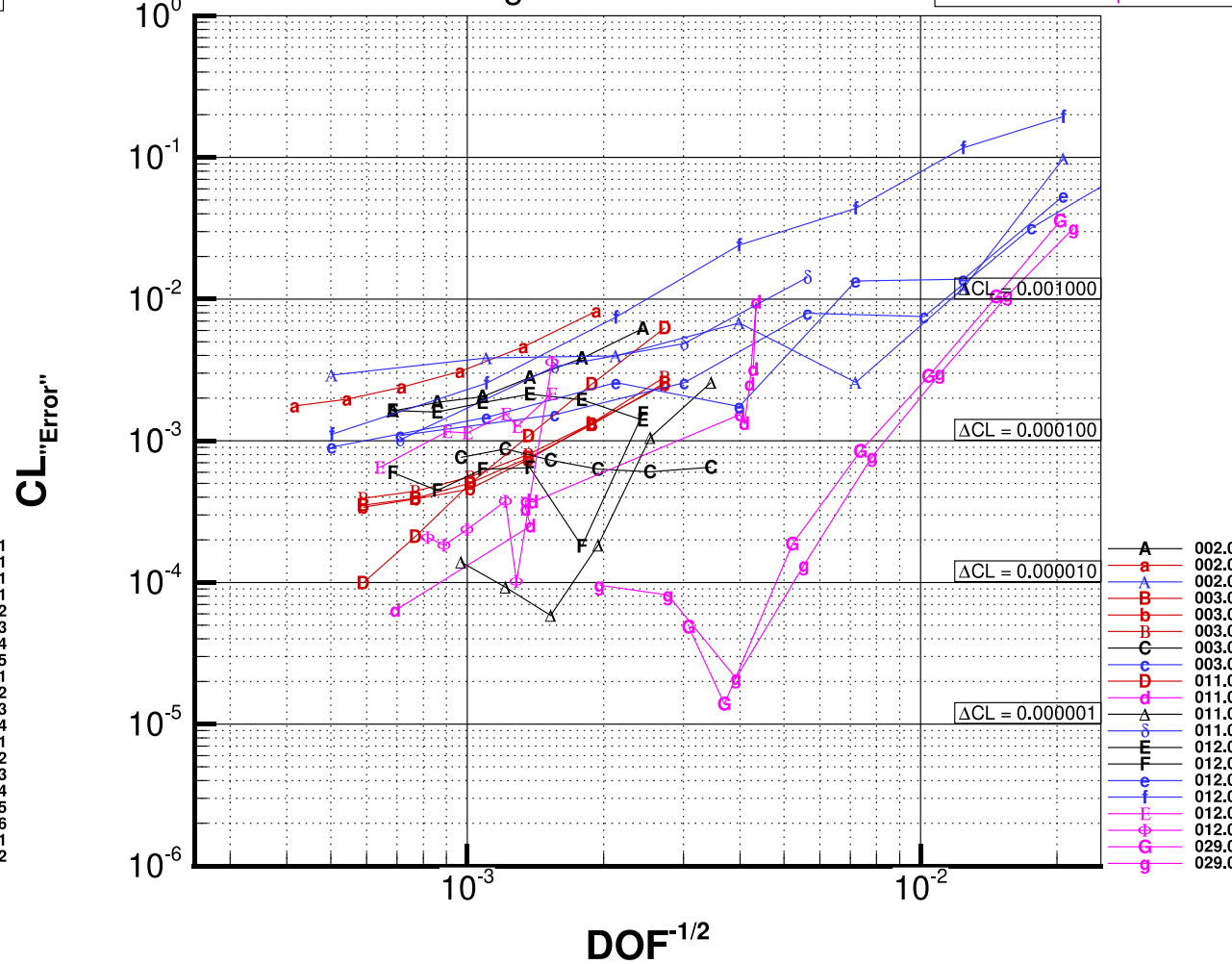
Test Case 1c: ONERA OAT15A
Grid Convergence Study
Angle of Attack 1.50

Grid Family:
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HeldenMesh Unstructured
User Grid Adaptation

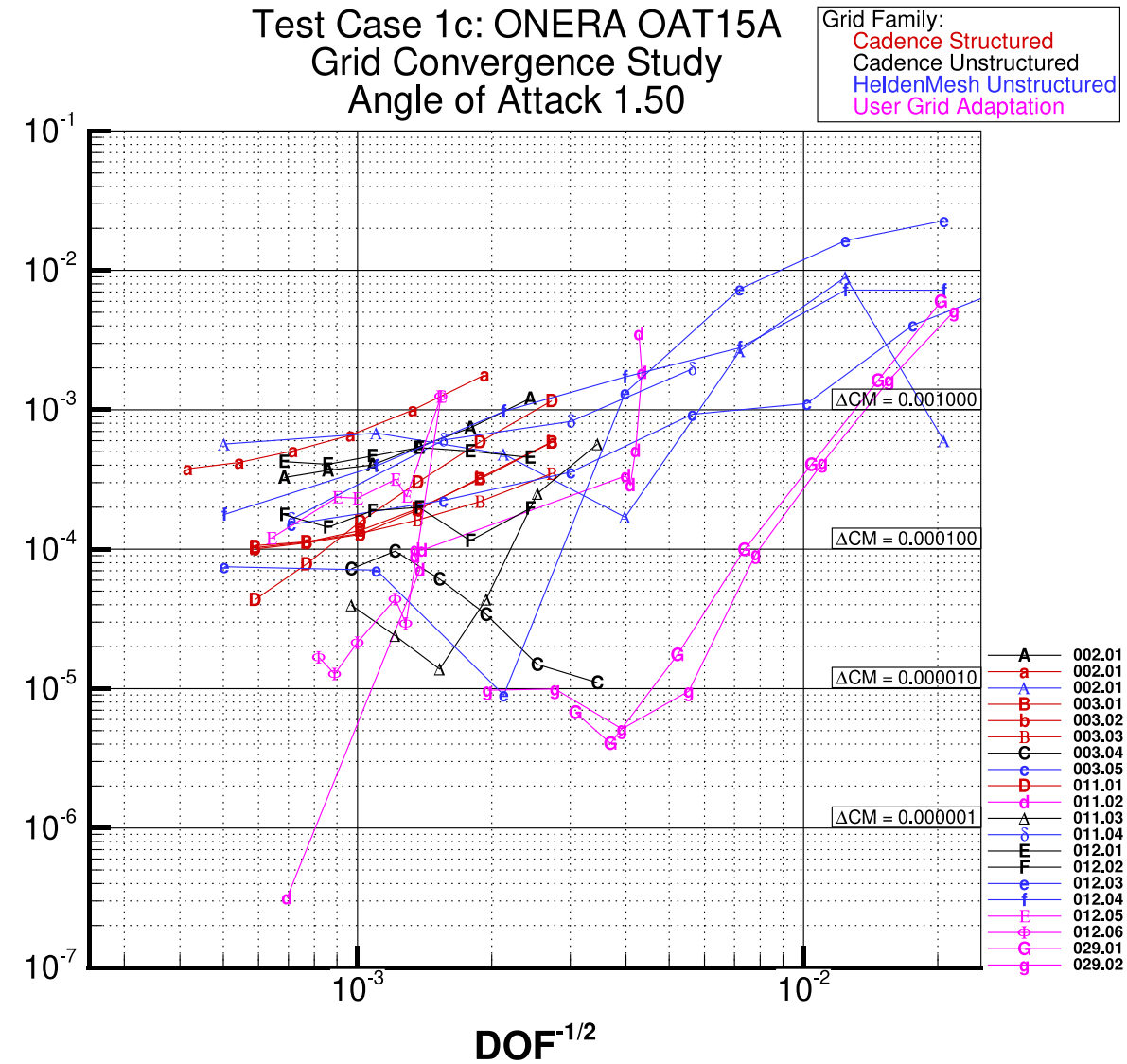
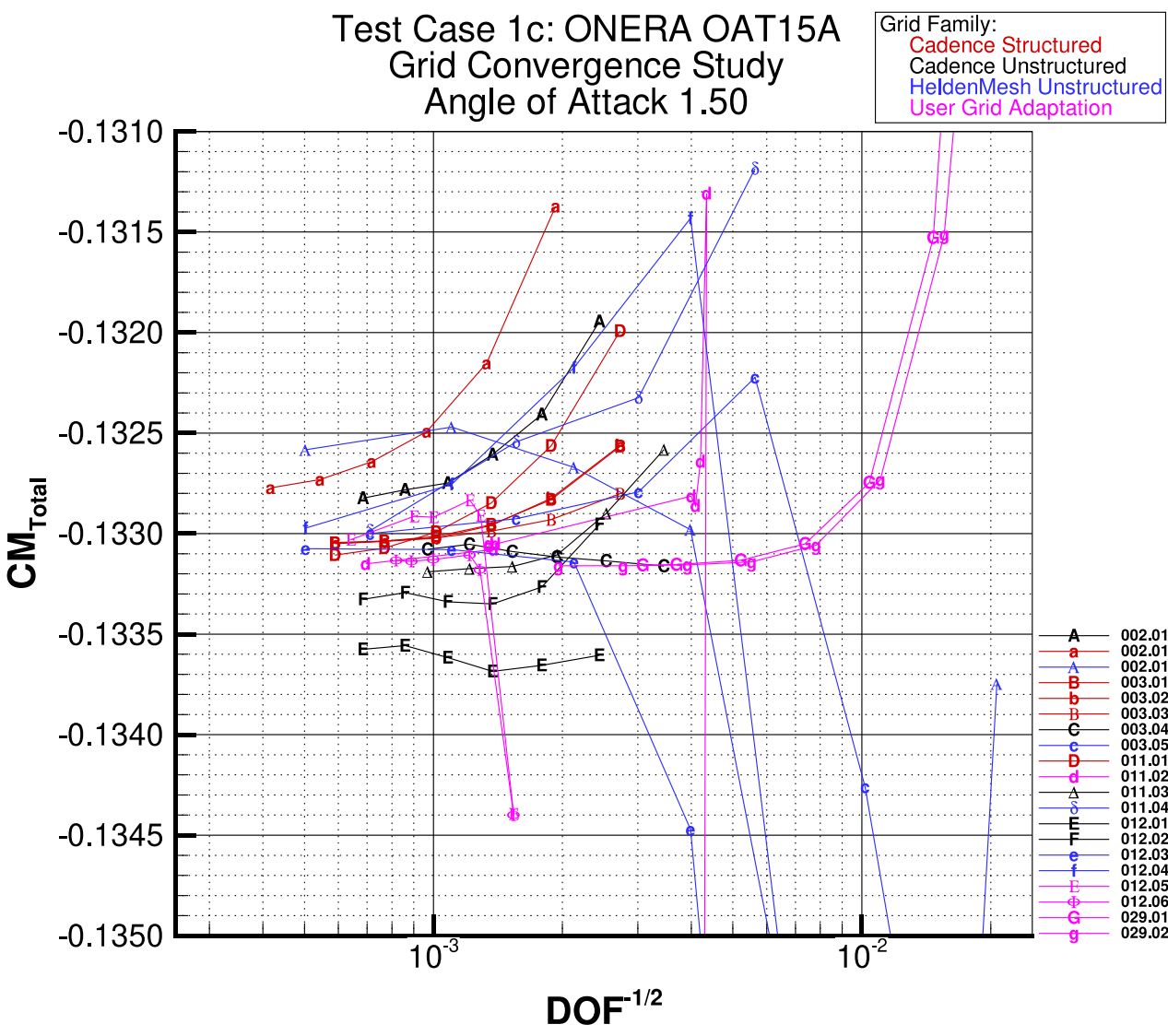


Test Case 1c: ONERA OAT15A
Grid Convergence Study
Angle of Attack 1.50

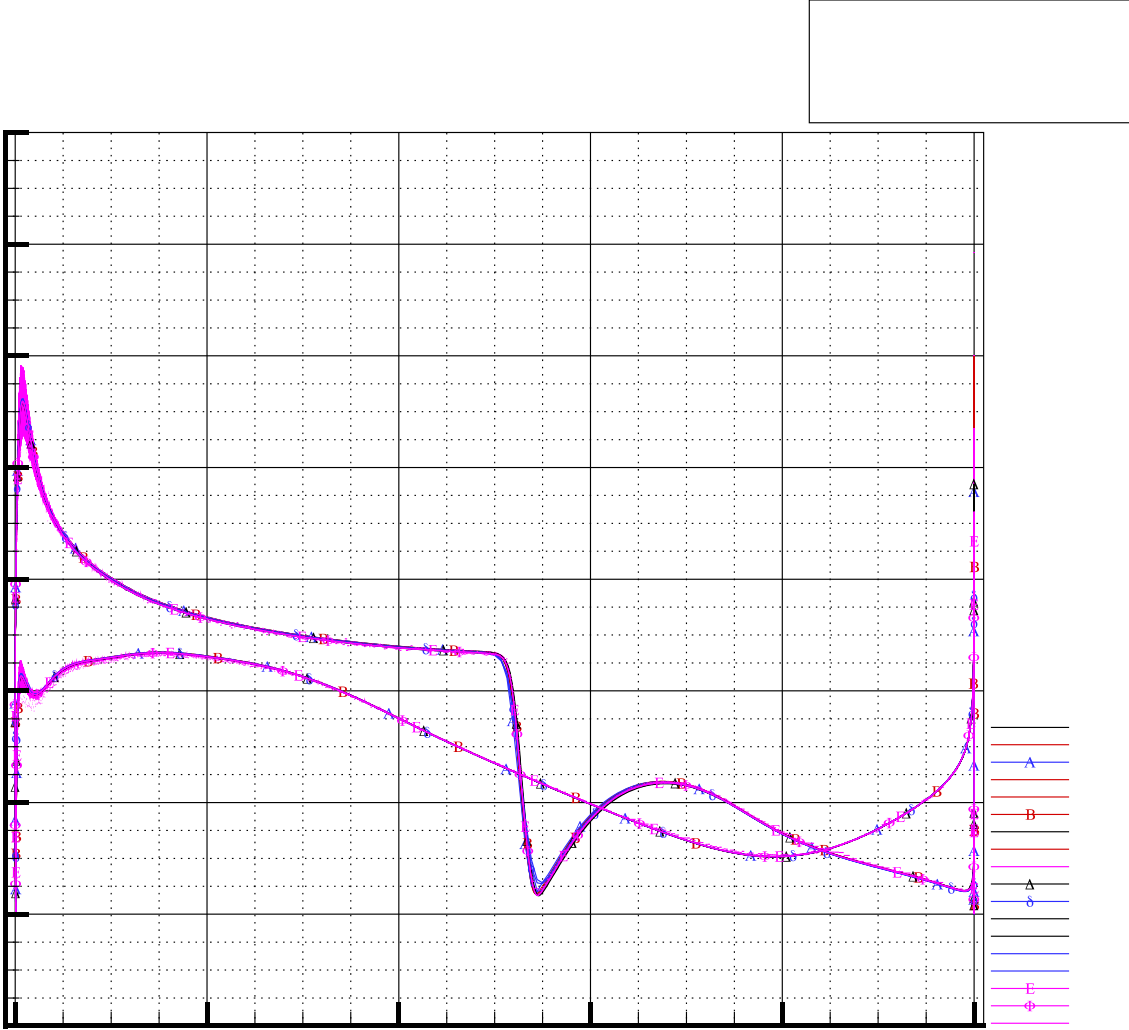
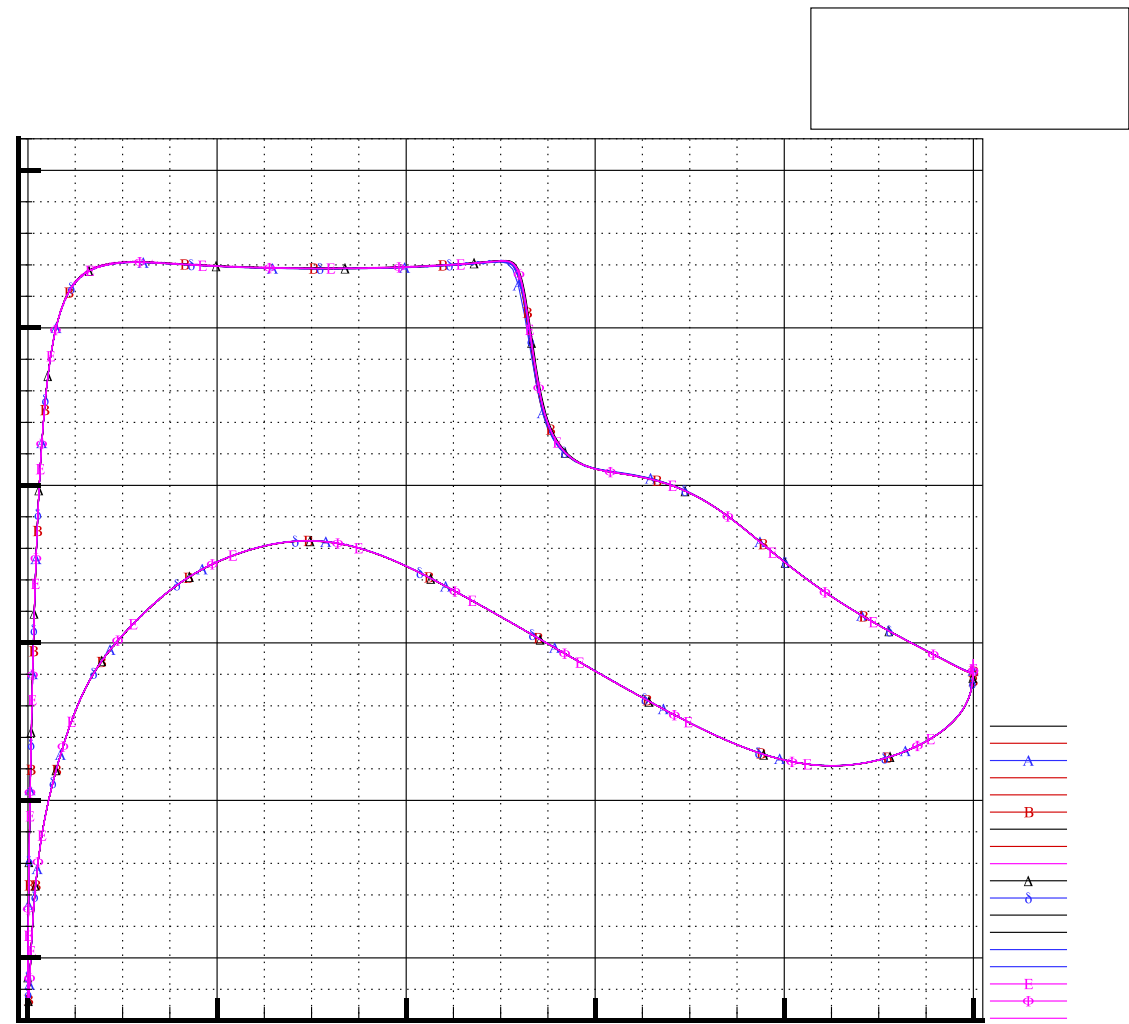
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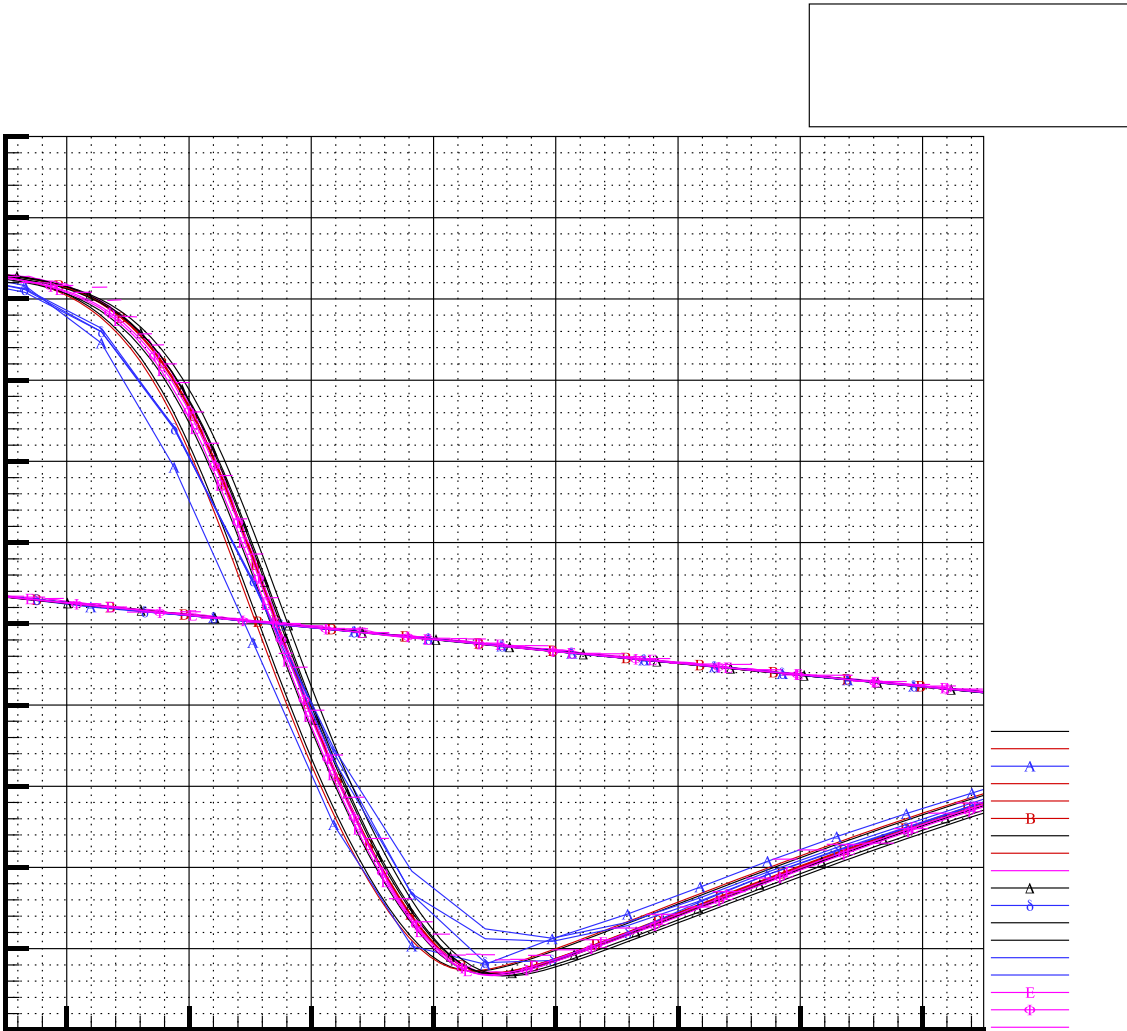
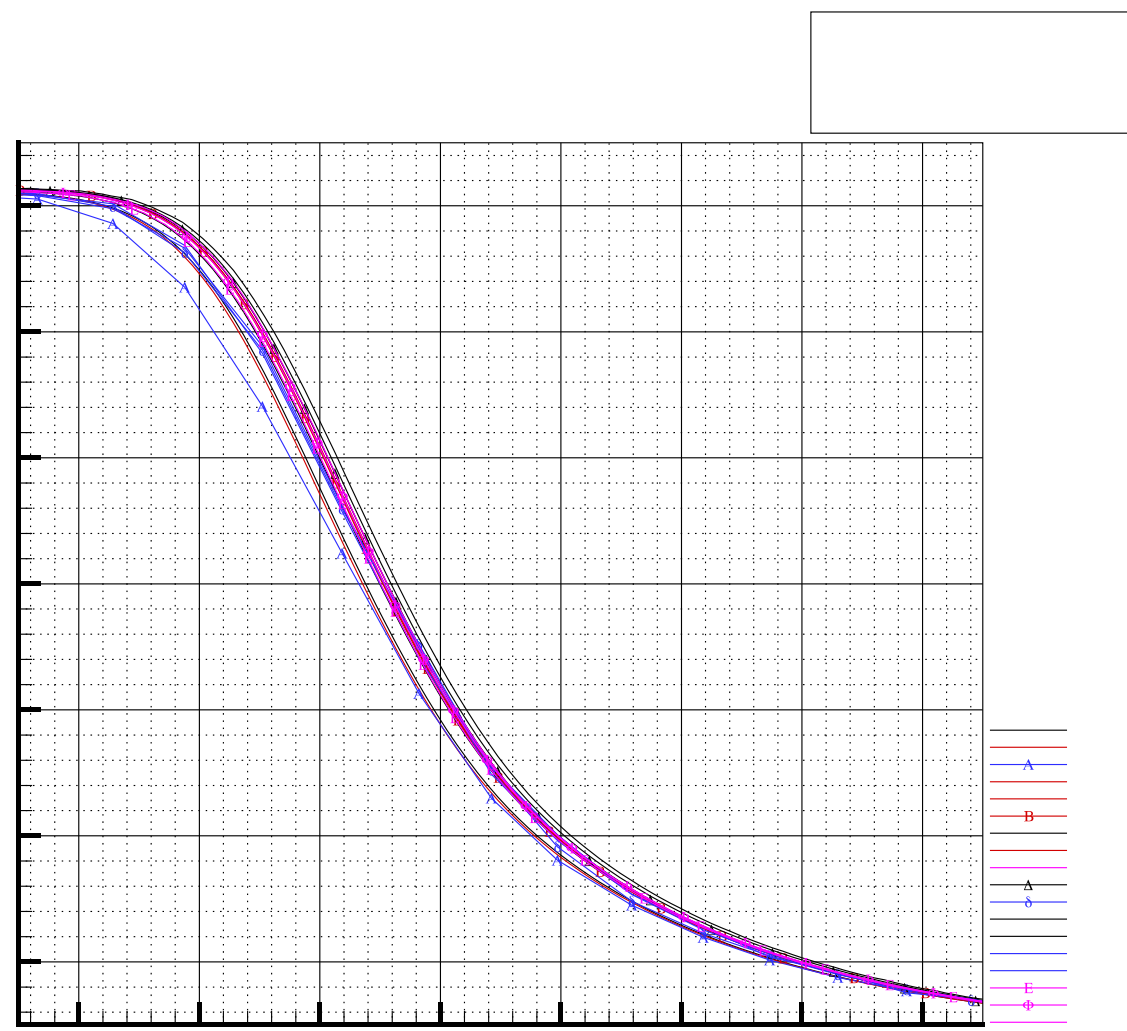
ONERA OAT15A C_M Convergence: $\alpha = 1.5^\circ$



ONERA OAT15A C_p and C_f $\alpha = 1.5^\circ$



ONERA OAT15A C_p and C_f $\alpha = 1.5^\circ$ (Shock)



Test Case 2a: Wing/Body Cruise

- Verification of steady CFD analysis, required
- Settings
 - Steady CFD RANS **French Vanilla SA-[neg]** vs **QCR2000 (All terms!)**
 - Adiabatic Wall (not isothermal)
 - Converge residuals to machine precision (~1e-10)
- Grids: https://www.aiaa-dpw.org/DPW8/Scatter/Test_Case_2
 - NASA CRM geometry including deformed wing matching condition
 - (L1:Itiny/L2:Coarse/L3:Medium/L4:Fine/L5:eXtra-fine/L6:Ultra-fine)
 - Six-member grid family; four are required, six are desirable
 - Encourage use of committee-supplied grids; user-generated grids are acceptable

Comparison Data

NTF197: r44,r51,r53
NTF215: r43,r103
NTF229: r296,r300,r302
Ames216: r35,r126,r130,r133

Reference Units

Sref (semi-span grid)	Cref	Semispan	Moment Center
297360.0 sq.in	278.5 in	1156.75 in	(1325.90, 0.00, 177.95)

Conditions

Mach	Re _c	α	T _{static} (120° F)	γ	Pr	Pr _t	Farfield $\chi = \tilde{v}/v$
0.85	5×10^6	2.50°	579.67 R 322.04 K	1.4	0.72	0.90	3

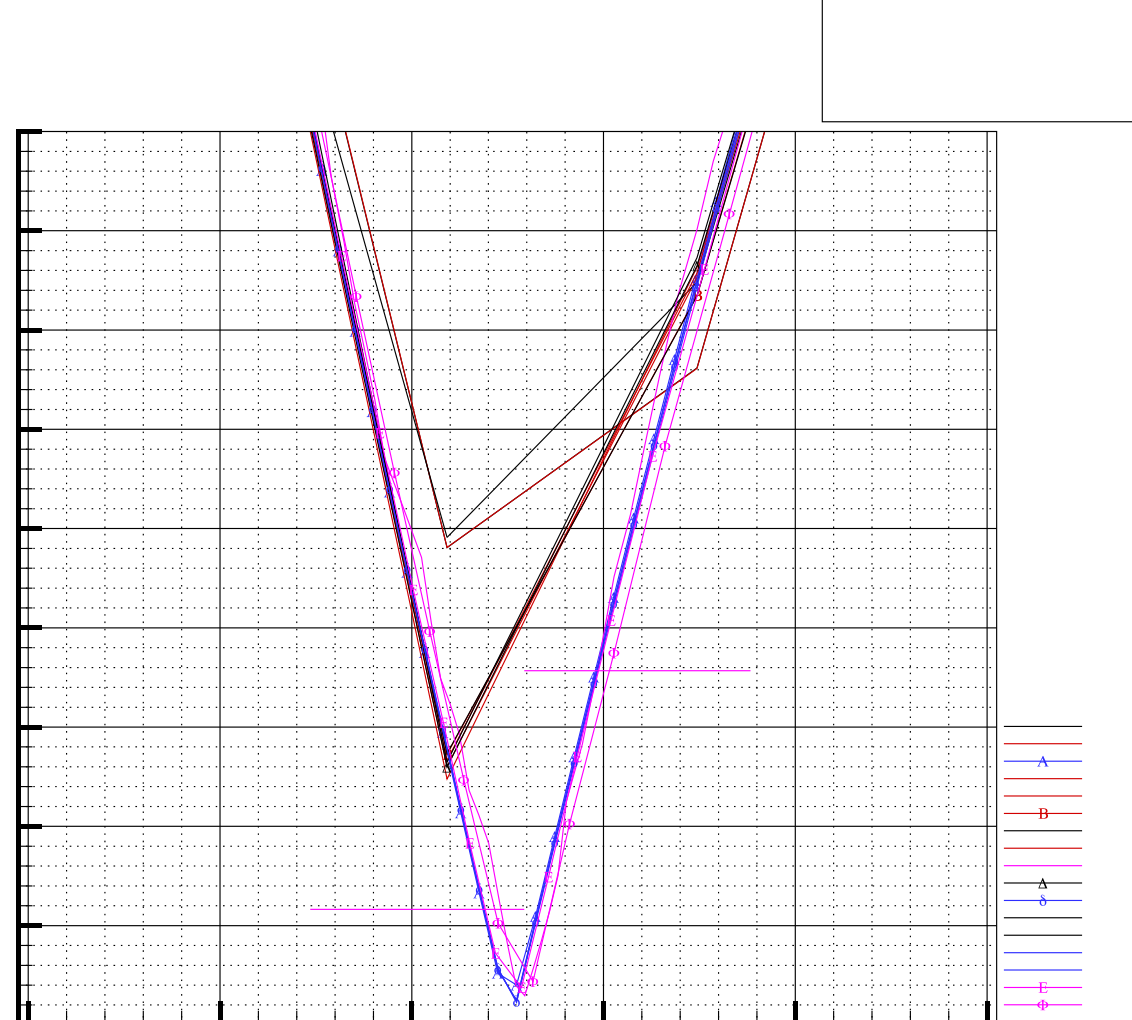
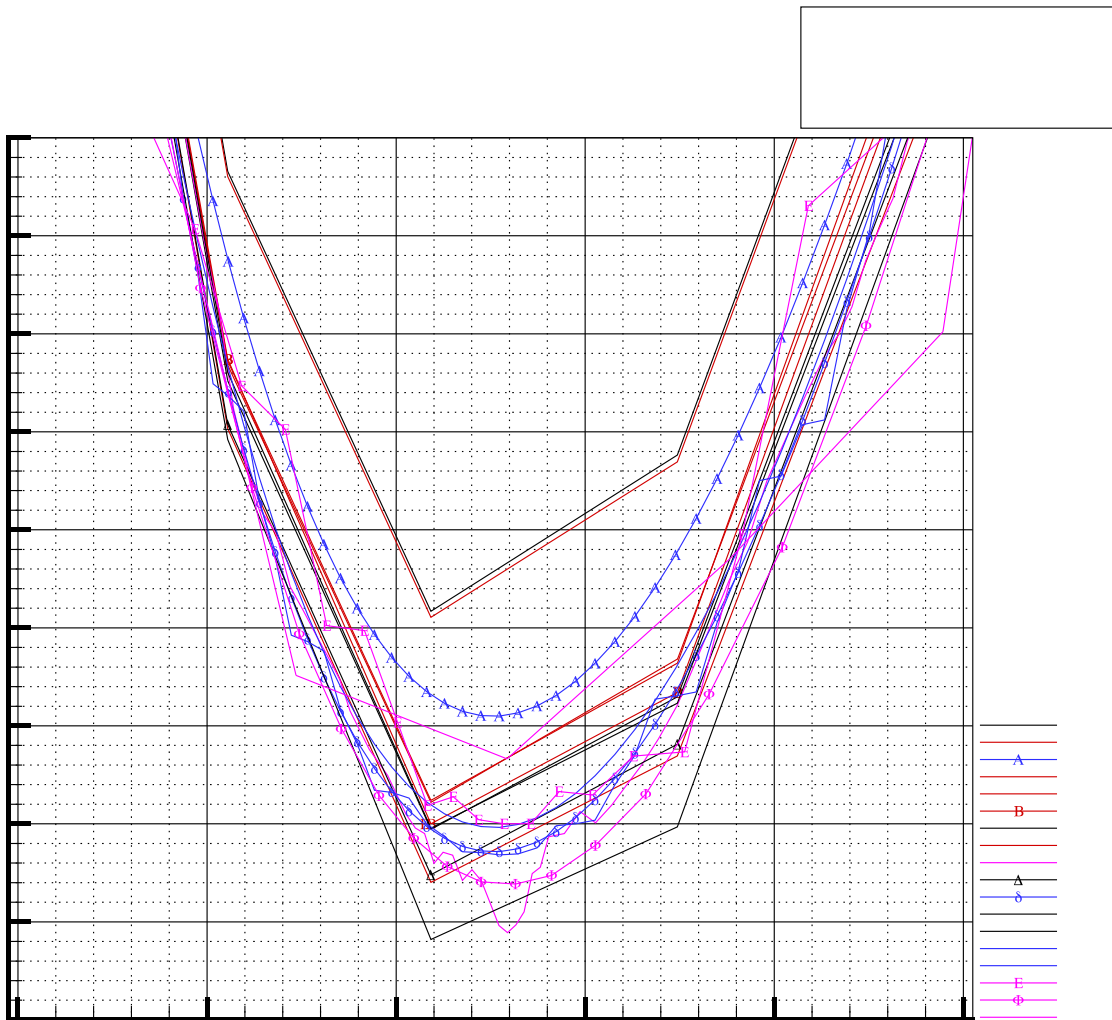
Sutherland's Law

$$\mu(T) = \mu_0 \left(\frac{T}{T_0} \right)^{3/2} \left(\frac{T_0 + S}{T + S} \right) \quad \mu_0 = 1.716 \times 10^{-5} \frac{\text{kg}}{\text{m s}} \quad T_0 = 491.6^\circ \text{R} \quad S = 198.6^\circ \text{R} \quad \frac{\mu(T)}{\mu_{ref}} = \left(\frac{T}{T_{ref}} \right)^{3/2} \left(\frac{1 + S/T_{fef}}{T/T_{fef} + S/T_{fef}} \right)$$

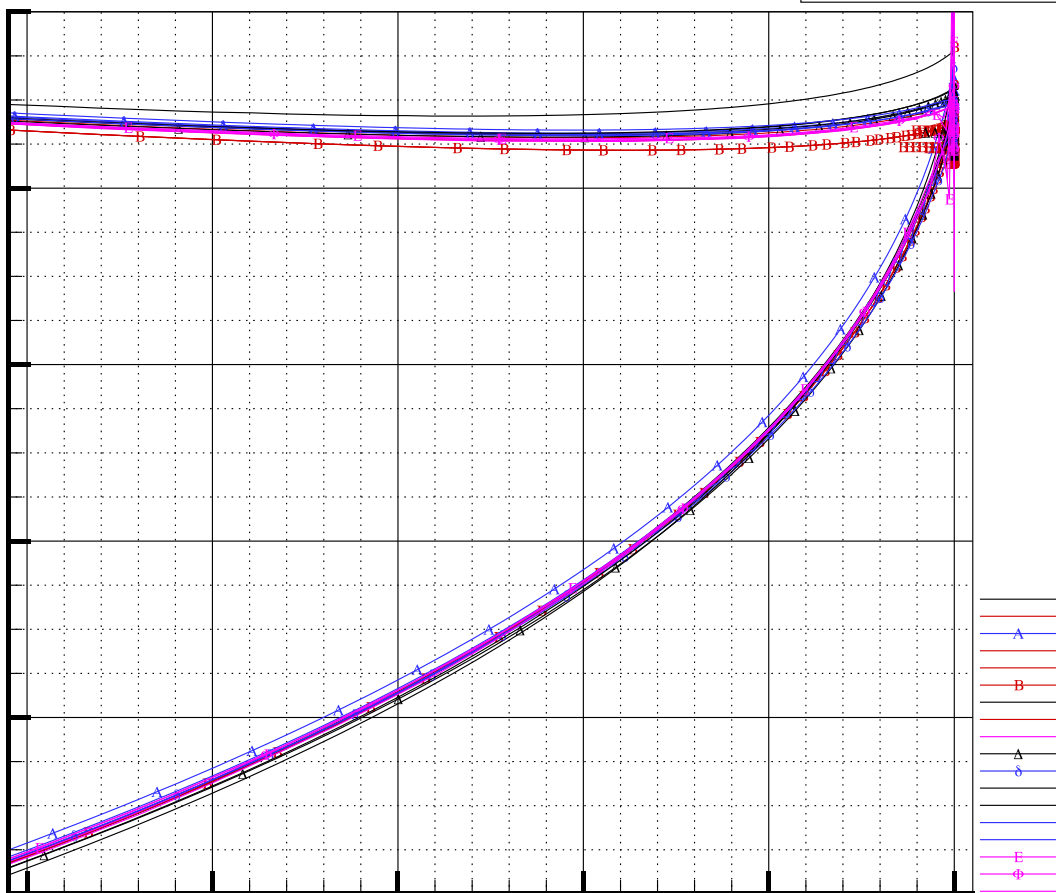
- **Test Case 1a: ONERA OAT15A**
 - Surprisingly large scatter!
- **Test Case 1b: Joukowski**
 - Excellent agreement between participants
 - Clear demonstration of 2nd-order accuracy (and higher!)
- **Test Case 1c: ONERA OAT15A**
 - Reduction in scatter!
 - Consistent Turbulence Model
 - Consistent problem definition (nothing left to participants)
 - Fairfield Distance
 - Residual Convergence
- **Test Case 2a: Wing/Body Cruise**
 - SA-[neg] vs SA-[neg]-QCR2000
- **Want to Participate?**
 - Contact galbramc@mit.edu or Ben.J.Rider2@boeing.com



ONERA OAT15A C_p and C_f $\alpha = 1.5^\circ$ (Leading Edge)



ONERA OAT15A C_p and C_f $\alpha = 1.5^\circ$ (Trailing Edge)



Potential Future Test Cases

Test Case 1d: ONERA OAT15A Airfoil

- **Verification of steady CFD analysis, required**

- **Settings**

- Steady CFD RANS **French Vanilla SA-[neg] + QCR2000**
 - Adiabatic Wall (not isothermal)
 - Characteristic Farfield (**1000 chords away**)
 - Use periodic boundary conditions for sidewall boundary conditions
- Converge residuals to machine precision ($\sim 1e-10$)

- **Grids**

- Six-member grid family; four are required, six are desirable
- Encourage use of committee-supplied grids; user-generated grids are acceptable

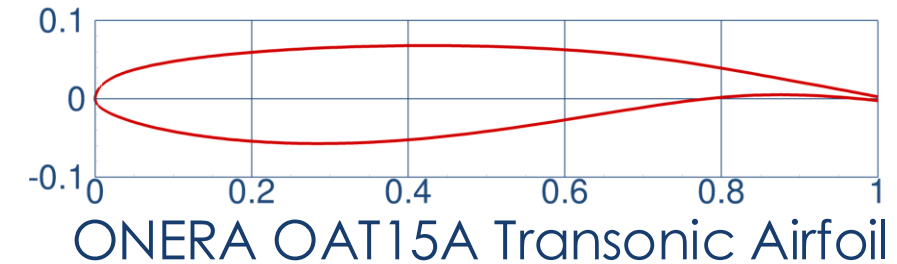
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0.73	3×10^6	271 K (487.8 R)	1.5°	1.4	0.72	0.9	3

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Test Case 2b: Wing/Body Cruise



- Verification of steady CFD analysis, required
- Settings
 - Steady CFD RANS **French Vanilla SA-[neg] + QCR2000 (All terms!)**
 - Adiabatic Wall (not isothermal)
 - Converge residuals to machine precision (~1e-10)
- Grids: https://dpw.larc.nasa.gov/DPW8/Scatter/Test_Case_2
 - NASA CRM geometry including deformed wing matching condition
 - (L1:Itiny/L2:Coarse/L3:Medium/L4:Fine/L5:eXtra-fine/L6:Ultra-fine)
 - Six-member grid family; four are required, six are desirable
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Ames216: r35,r126,r130,r133

Reference Units

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Conditions

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0.85	5 × 10 ⁶	2.50°	579.67 R 322.04 K	1.4	0.72	0.90	3

Sutherland's Law

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Test Case 2c: Wing/Body Cruise



- Far Field Sensitivity Study: Steady CFD analysis
- Settings
 - Steady CFD RANS **French Vanilla SA-[neg] + QCR2000 (All terms!)**
 - Adiabatic Wall (not isothermal)
 - Converge residuals to machine precision (~1e-10)
- Grids: https://dpw.larc.nasa.gov/DPW8/Scatter/Test_Case_2
 - NASA CRM geometry including deformed wing matching condition
 - (L1:Itiny/L2:Coarse/L3:Medium/L4:Fine/L5:eXtra-fine/L6:Ultra-fine)
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Test Case 3a: Wing/Body Polar

- **CRM Wing/Body**

- Reynolds number: 5M (LoQ)
- Dynamic Pressure: $Q_{\infty} = 1384$ psf
- Mach number: 0.85 (M_{cruise})
- Angles of attack: 2.50, 3.00, 3.25, 3.50, 4.00, 4.25
- Temperature: 120.0 F (579.67 R / 322.04 K)
- Reference Information: <https://aiaa-dpw.larc.nasa.gov/Workshop7/DPW7-geom.html>

Grid: Level 3
Grid: Level 1-6

Comparison Data

NTF197: r44,r51,r53
NTF197: r92,r97,r99 (WBT0)
NTF215: r43,r103
NTF229: r296,r300,r302
Ames216: r35,r126,r130,r133

- **Committee-supplied**

- NASA CRM geometry including deformed wing matching condition
 - Trip location – Wing: 10% chord upper/lower surface
- Grid Family (L1:Tiny/L2:Coarse/L3:Medium/L4:Fine/L5:eXtra-fine/L6:Ultra-fine)

- **Comparison metrics**

- Forces / Moments
- Sectional Twist / Deformation
- Sectional C_p distribution
- Residuals (Flow & Structural Solver)

Test Case 3b: Wing/Body Polar (R=20M)

- **CRM Wing/Body**

- Reynolds number: 20M (HiQ)
- Dynamic Pressure: $Q_\infty = ?$
- Mach number: 0.85 (M_{cruise})
- Angles of attack: -1.50, 0.00, 1.50, 2.75, 3.10, 3.50, 4.00, 4.50
- Temperature: ? F (? R / ? K)
- Reference Information: <https://aiaa-dpw.larc.nasa.gov/Workshop7/DPW7-geom.html>

Grid: Level 3
Grid: Level 1-6

Comparison Data

NTF197: r?, r?
NTF197: r?, r? (WBT0)
NTF215: r?
NTF229: r?
ETW ESWIRP: r?

- **Committee-supplied**

- NASA CRM geometry including deformed wing matching condition
 - Trip location – Wing: 10% chord upper/lower surface
- Grid Family (L1:Tiny/L2:Coarse/L3:Medium/L4:Fine/L5:eXtra-fine/L6:Ultra-fine)

- **Comparison metrics**

- Forces / Moments
- Sectional C_p distribution
- Sectional Twist / Deformation
- Residuals (Flow & Structural Solver)

Test Case 3c: Wing/Body Mach Sweep

- **CRM Wing/Body**

- Reynolds number: 5M (LoQ)
- Dynamic Pressure: $Q_{\infty} = 2,221,384, 2 \text{ psf}$
- Mach numbers: 0.70, 0.80, 0.82, 0.85, 0.87
- Angle of attack: 2.75
- Temperature: $2, 2, 2, 120.0, 2 \text{ F}$ (579.67 R / 322.04 K)
- Reference Information: <https://aiaa-dpw.larc.nasa.gov/Workshop7/DPW7-geom.html>

Grid: Level 1-6

Comparison Data

NTF197: r?, r?
NTF197: r?, r? (WBT0)
NTF215: r?
NTF229: r?
ETW ESWIRP: r?
Ames216: r?

- **Committee-supplied**

- NASA CRM geometry including deformed wing matching condition
 - Trip location – Wing: 10% chord upper/lower surface
- Grid Family (L1:Tiny/L2:Coarse/L3:Medium/L4:Fine/L5:eXtra-fine/L6:Ultra-fine)

- **Comparison metrics**

- Forces / Moments
- Sectional Twist / Deformation
- Sectional C_p distribution
- Residuals (Flow & Structural Solver)

Scatter Reduction Working Group Leadership

- **Marshall Galbraith, Massachusetts Institute of Technology**
- **Kevin Holst, University of Tennessee, Knoxville**
- **Ben Rider, The Boeing Company**

Geometry/Grid Far Field Details

NASA CRM (Full Scale)

MAC = 275.8"

Semispan = 1156.75"



DPW8

DPW7: iges files (box)	Min	Max	+/- Range	Range / MAC	Range / Semispan
X (inch)	-22440.9	23838.6	23139.75	~ 84.0	~ 20.0
Y (inch)	0.0	35433.1	35433.1	~ 128.6	~ 30.6
Z (inch)	-23405.5	23838.6	23838.6	~ 86.5	~ 20.6

https://aiaa-dpw.larc.nasa.gov/Workshop7/Geometry/2021-03-02_Version_01/DPW7geometries.zip

DPW8: HeldenMesh (box)	Min	Max	+/- Range	Range / MAC	Range / Semispan
X (inch)	-55160.0	55160.0	55160.0	200.0	~ 47.7
Y (inch)	0.0	55160.0	55160.0	200.0	~ 47.7
Z (inch)	-55160.0	55160.0	55160.0	200.0	~ 47.7

https://dpw.larc.nasa.gov/DPW8/Scatter/Test_Case_2/Helden_Grids.REV00/

DPW8: Cadence (box)	Min	Max	+/- Range	Range / MAC	Range / Semispan
X (inch)	-115582.5	118237.9	116910.2	~ 424.2	~ 101.1
Y (inch)	0.0	116834.7	116834.7	~ 423.9	~ 101.0
Z (inch)	-115584.0	116018.1	115801.1	~ 420.2	~ 100.1

https://dpw.larc.nasa.gov/DPW8/Scatter/Test_Case_2/Cadence_Grids.REV00/

DPW7

DPW7: Vassberg (sphere)	Min	Max	+/- Range	Range / MAC	Range / Semispan
X (inch)	-30328.203	32996.560	31662.382	~ 114.8	~ 27.4
Y (inch)	0.0	31664.519	31664.519	~ 114.8	~ 27.4
Z (inch)	-31449.638	31865.992	31657.815	~ 114.8	~ 27.4

https://dpw.larc.nasa.gov/DPW7/Vassberg_Grids.REV00/

DPW7: JAXA (sphere)	Min	Max	+/- Range	Range / MAC	Range / Semispan
X (MAC)	-413.890	424.690	419.290	~ 419.3	~ 100
Y (MAC)	0.0	419.349	419.349	~ 419.3	~ 100
Z (MAC)	-418.675	420.005	419.340	~ 419.3	~ 100

https://dpw.larc.nasa.gov/DPW7/JAXA_Grids.REV00/

DPW7: NLR (sphere)	Min	Max	+/- Range	Range / MAC	Range / Semispan
X (m)	-2949.98	3050.00	2999.99	~ 428.2	~ 102.1
Y (m)	0.0	3000.00	3000.00	~ 428.2	~ 102.1
Z (m)	-2995.1	3004.99	3000.05	~ 428.2	~ 102.1

https://dpw.larc.nasa.gov/DPW7/NLR_Grids.REV00/DPW7-NLR-grids/

DPW7: DLR (box)	Min	Max	+/- Range	Range / MAC	Range / Semispan
X (m)	-570.0	630.0	1200.0	~ 171.3	~ 40.9
Y (m)	0.0	900.0	900.0	~ 128.5	~ 30.6
Z (m)	-594.5	605.5	1200.0	~ 171.3	~ 40.9

https://dpw.larc.nasa.gov/DPW7/DLR_Grids.REV00/

DPW7: GGNS (box)	Min	Max	+/- Range	Range / MAC	Range / Semispan
X (inch)	-22400.0	22400.0	22400.0	~ 81.2	~ 19.4
Y (inch)	0.0	22400.0	22400.0	~ 81.2	~ 19.4
Z (inch)	-22400.0	22400.0	22400.0	~ 81.2	~ 19.4

Adapted grids (not publicly released)